PROTOTYPING

LOW- & HIGH-FIDELITY

What is Prototyping?

- An original or first model of something from which other forms are copied or developed. (Webster Dictionary)
- A prototype allows stakeholders to interact with an envisioned product, to gain some experience of using it in a realistic setting, and to explore imagined uses. (Interaction Design Book)
- A method for developing a model of a system that allows to test the approach early on.
- “You can fix it on the drafting board with an eraser, or you can fix it later with a sledgehammer.” (Frank Lloyd Wright)



Prototypes

- prototypes serve as **externalizations** or **manifestations** of design ideas
- they can be used to reflect about ideas, to communicate ideas, and to test ideas

The goal of a prototype is to test the flow of a design solution and gather feedback on it — from both internal and external parties — before constructing the final product

Why Prototype?

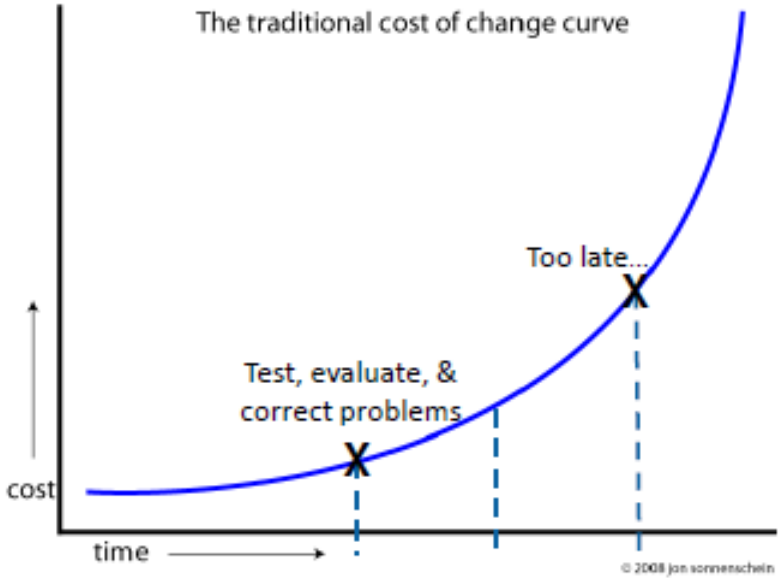
- allow you to **fail faster, learn sooner, and succeed earlier**
- allow you to **identify interaction fails**
- are useful when discussing or **evaluating ideas** with stakeholders
- encourage **early user participation** and involvement
- a **communication device among team members** and an effective way for designers to explore design ideas

Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019



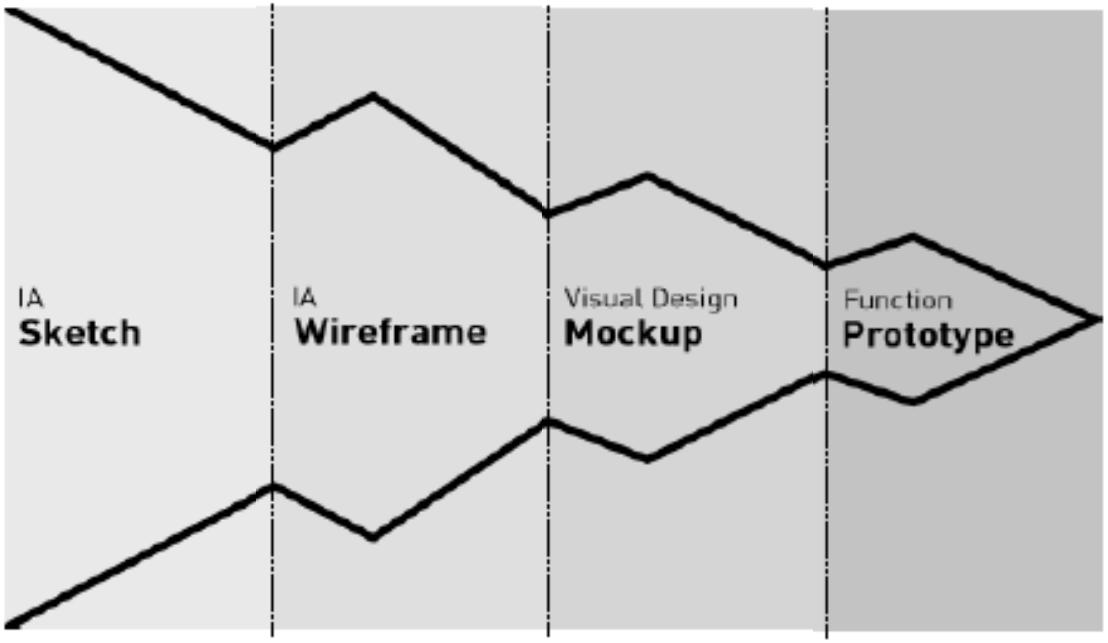
Why Prototype?

- support designers in **choosing between alternatives**
- using **feedback** for selection and further development
- **Specification** of vague requirements
- Allows to **deal with problems** before it is too late
- Multiple iterations allow to **reduce the risk** at market launch



Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019

Prototype – Exploration and Reduction



Concept

Explorate alternatives

Clarify

Refine

When Prototyping?

- Before the design process
  To present the applicability of the concept
- At the beginning of the design process
  To derive and test requirements
- During the design process
  Usability tests, “Early marketing” and demonstrations
- At the end of the design process
  Evaluate and generate new requirements

Potential problems of Prototyping

- Potentially insufficient analysis
- Users might not be satisfied if many features are not implemented
- Some might just “extend” a prototype to become the full product



Prototyping

Resolution: “Auflösung, Funktionalität”

- overall extent of the implementation of a prototype (e.g., only one screen in detail vs. the complete system as a rough overview).

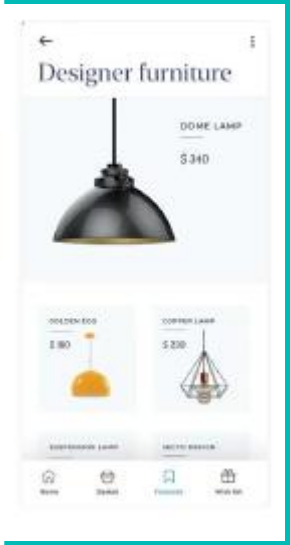
Fidelity: “Detailgenauigkeit”

- fidelity of rendering, accuracy of implementation (e.g., line drawing on paper, Illustrator sketch, or photorealistic rendering).

Fidelity of prototyps

Level of detail





Low fidelity

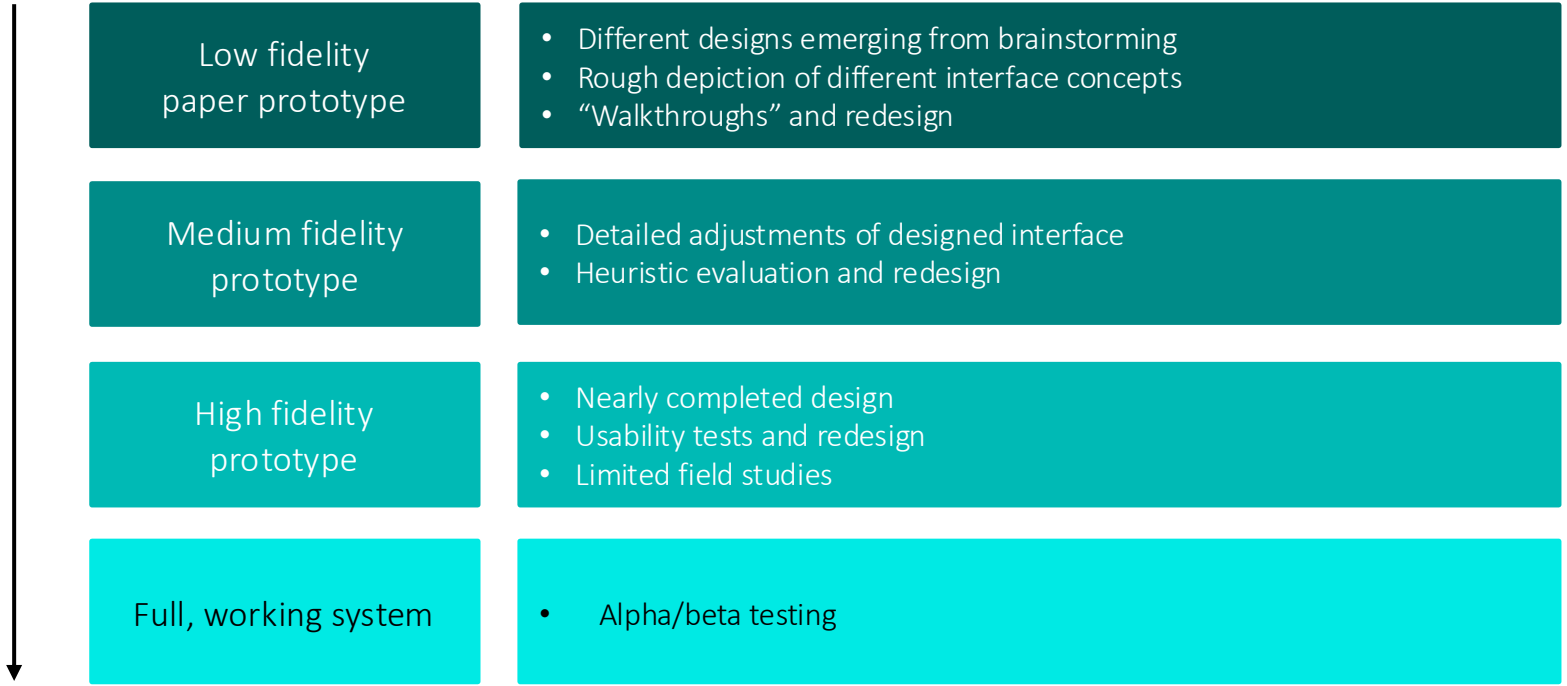
- less finished but more flexible
- less costs
- can be constructed more rapidly
- quick to modify
- repeated prototyping and testing enables rapid iteration and refinement of the design

High fidelity

- more detailed
- more time and effort for creation and modification
- closer to the final product (both in design and function)
- appropriate for mature design stage and when concepts need realism to be evaluated

Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019

Prototyping - Fidelity



Prototyping

- **Low resolution** because low extend of implementation
- **Low fidelity** because It does not “feel” like a real car in any way
- **For exploration only**
  - in early design stages
  - cheap, simple, quick to produce and modify
  - limited expressiveness



Prototyping

- **Low resolution** because low extend of implementation
- **High fidelity** because it looks like the real car
- Attention: this is only true for the question of “how will the car look?” “Horizontal Prototype”
- **Low fidelity** for
  - “how much space will it need”
  - “how will it feel to sit in it”
  - …



Prototyping

- **High resolution** because complete implementation

- **High fidelity** because experience of driving the car is as close as it gets

- Check details
- In late design stages
- Complete functionality
- Expensive, complex to build

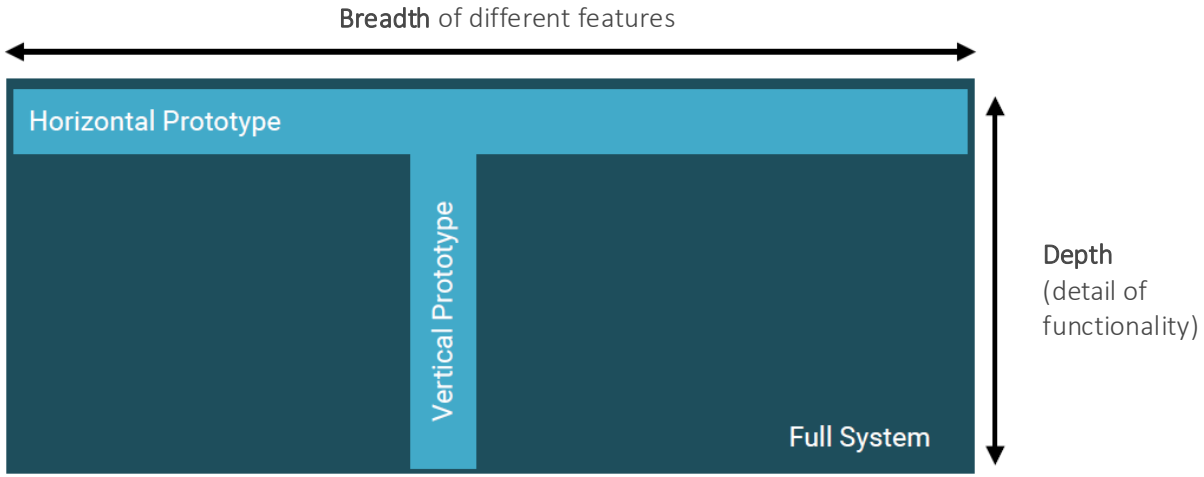


Types of Prototypes

- a prototype is a design representation that is in some way(s) less than a full implementation
- the choices for your approach to prototyping are about how to make it less

Horizontal prototype:
provides a wide range of functions but with little detail

Vertical prototype:
provides a lot of detail for only a few functions



Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019

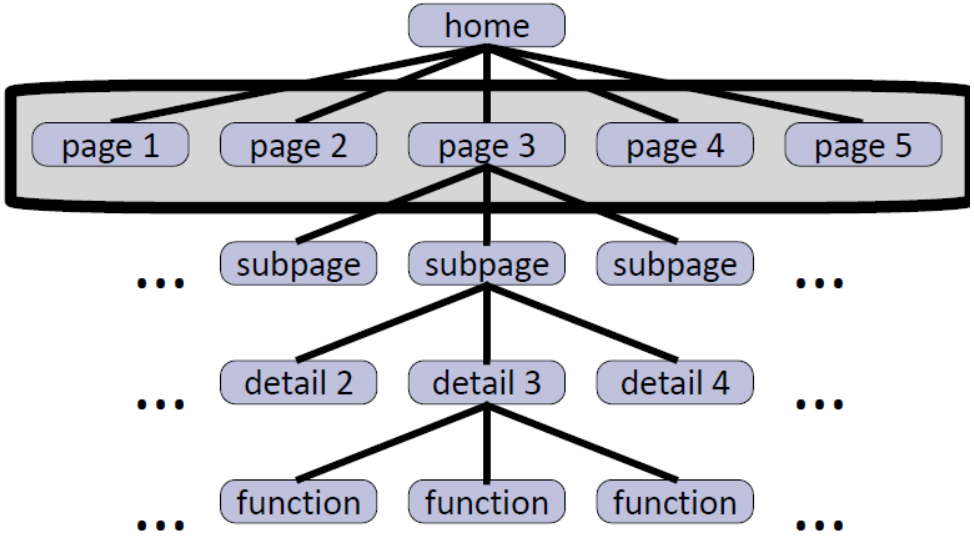
Horizontal Prototype

- Sketches feature spectrum of a product
- Entire UI visible, but no functionality
- Simulation of each interaction step

- Used in early design stages
  - to determine the set of features
  - to decide on user interface concept

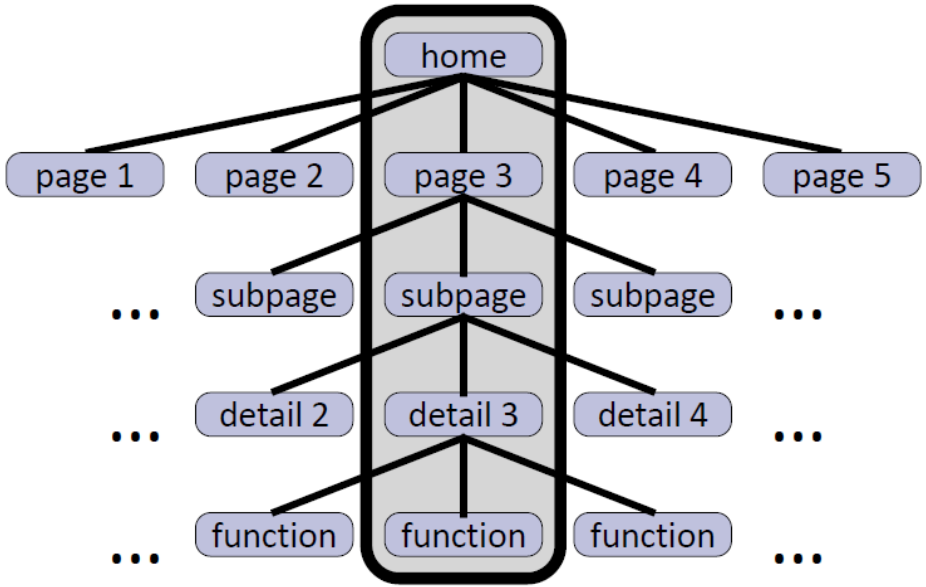
- Helps to evaluate
  - Navigation
  - User interface concept

- Example: Click dummy



Vertical Prototype

- Demonstrates a selected feature of a product
  - Compare two different realizations
- Few functions, but implemented in detail
- Testing general design ideas by example
- Used in later design stages
  - Optimize usage of a function
- Example: Proof-of-concept

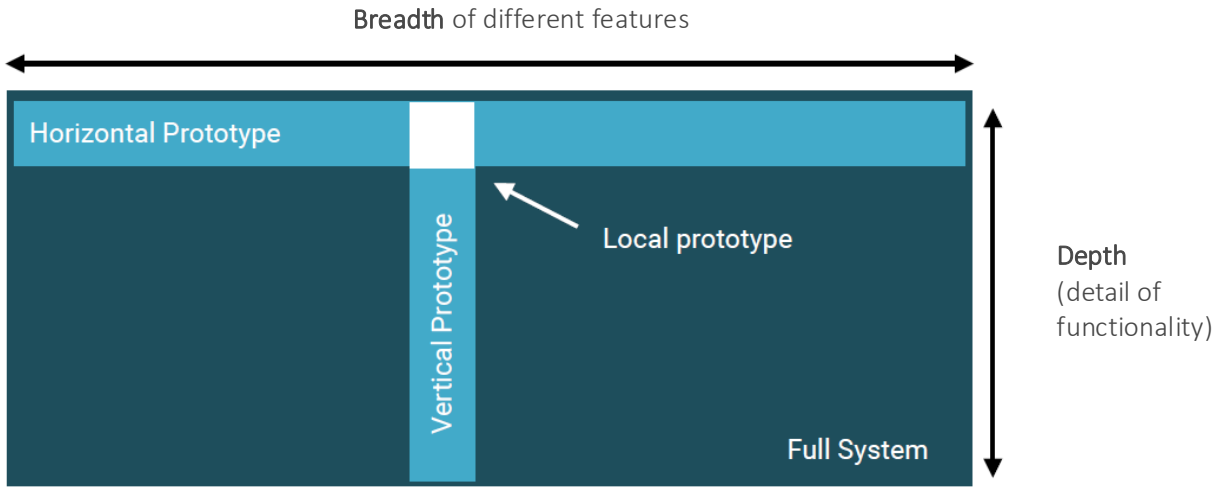


Types of Prototypes

- a prototype is a design representation that is in some way(s) less than a full implementation
- the choices for your approach to prototyping are about how to make it less

Local prototype:
narrow in both dimensions, limiting its focus to a localized interaction design issue

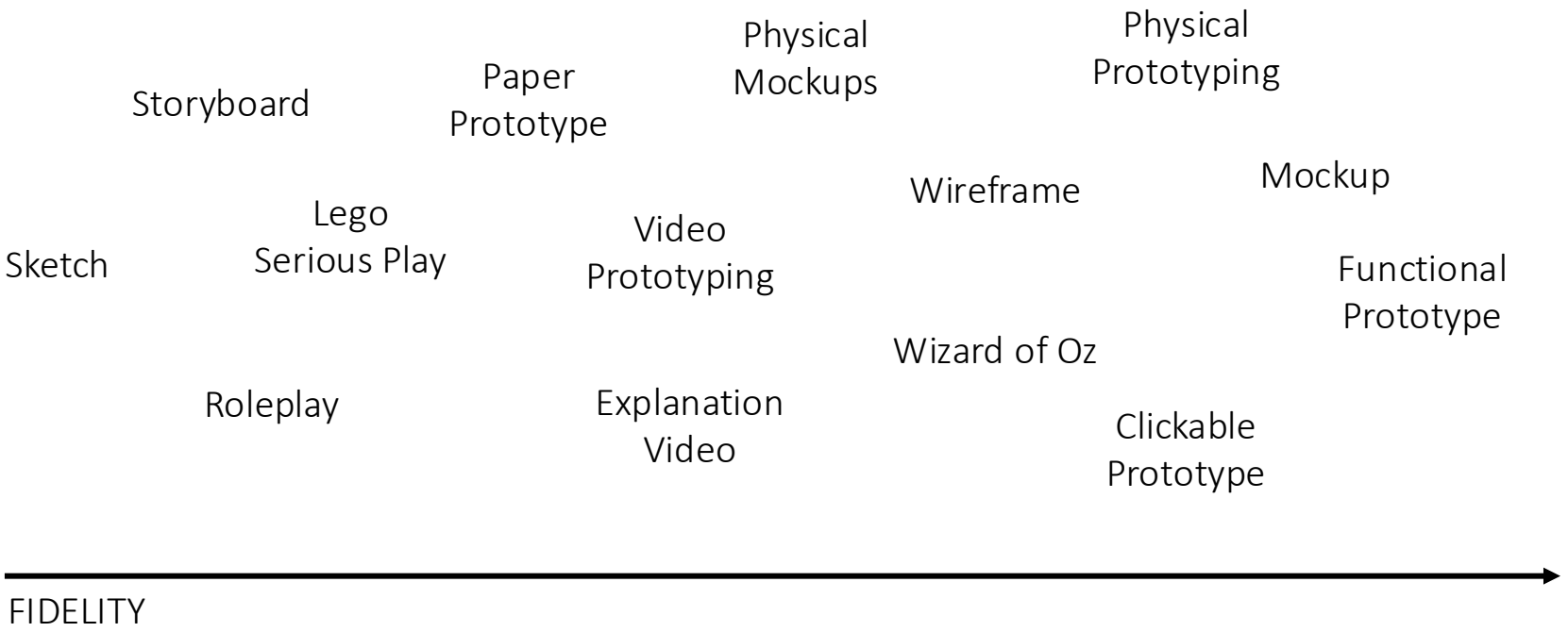
"T" prototype:
combines advantages of both horizontal and vertical prototypes



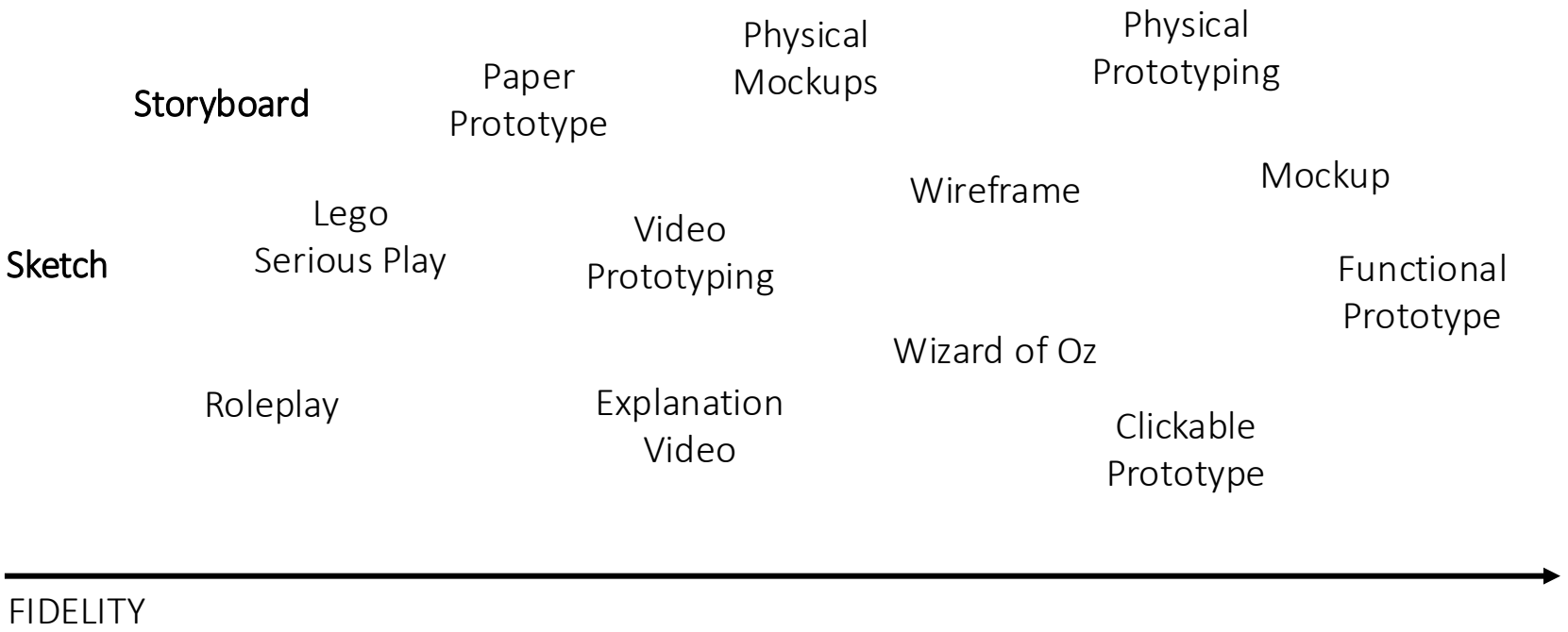
Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019

Don‘t waste time by coding too early!

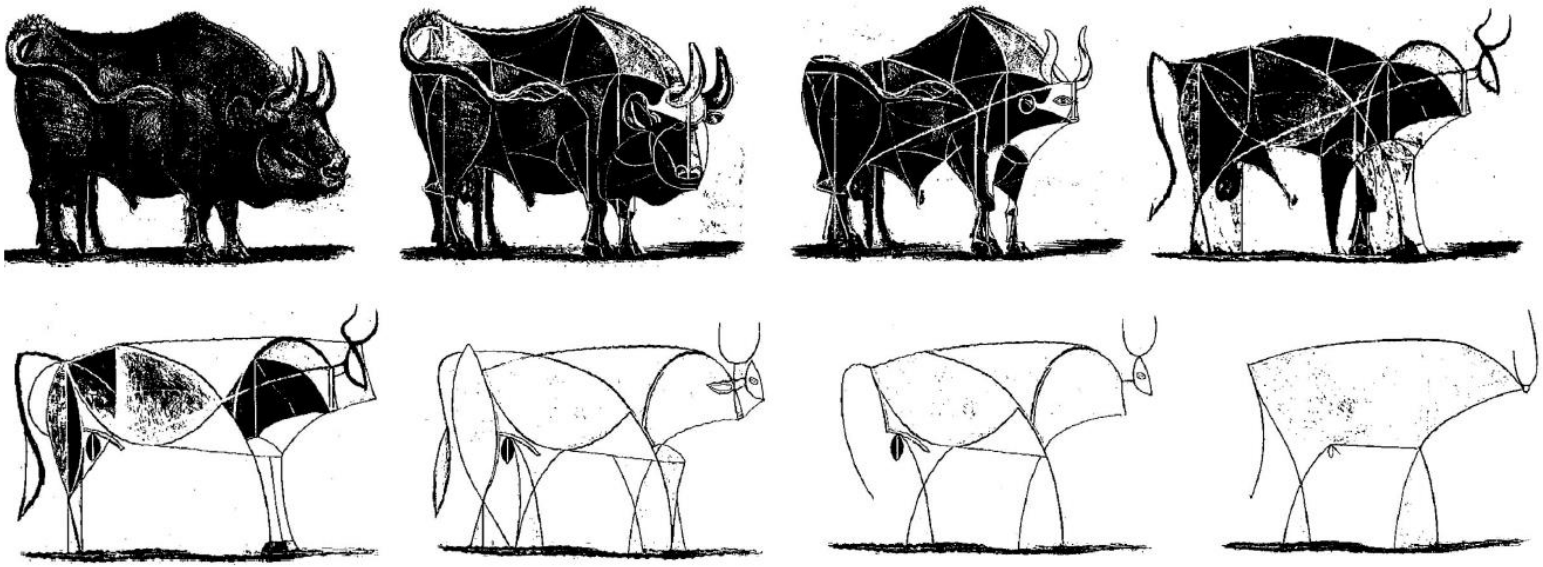
Prototyping Methods



Prototyping Methods



Sketching & Storyboard



Pablo Picasso - Series of lithographs on the “Abstraction of a Bull”

Sketching & Storyboard

are used to visualize ideas or concepts
purpose: think through ideas and get agreement from the team and stakeholders

- **Sketch** = single scene (idea, concept, interface etc.)
- **Storyboard** = sequence of visual frames that illustrate the interaction between the persona and the intended device or environment



helpful to view the interaction in context

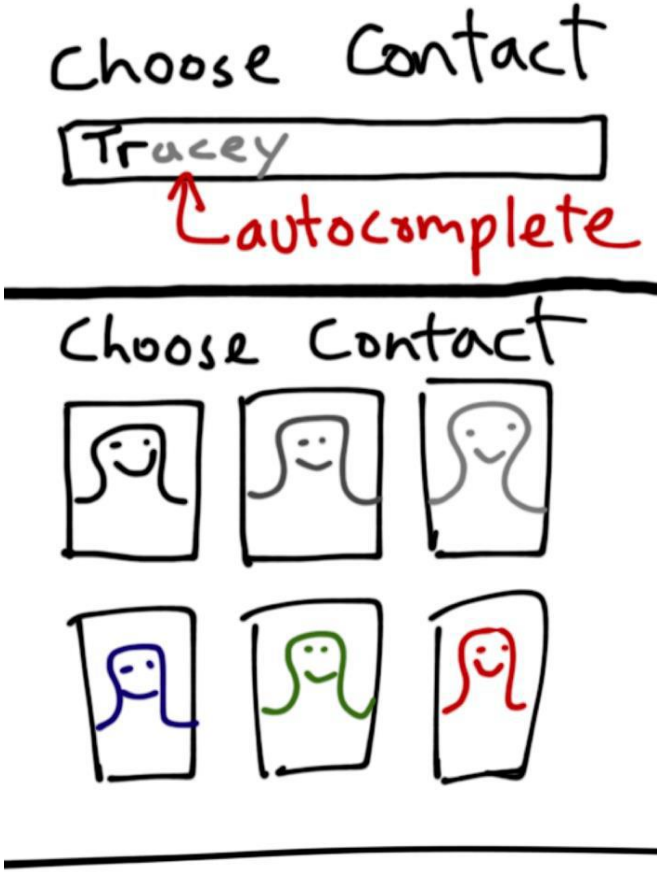
Greenberg et al., 2011. Sketching User Experiences: The Workbook.

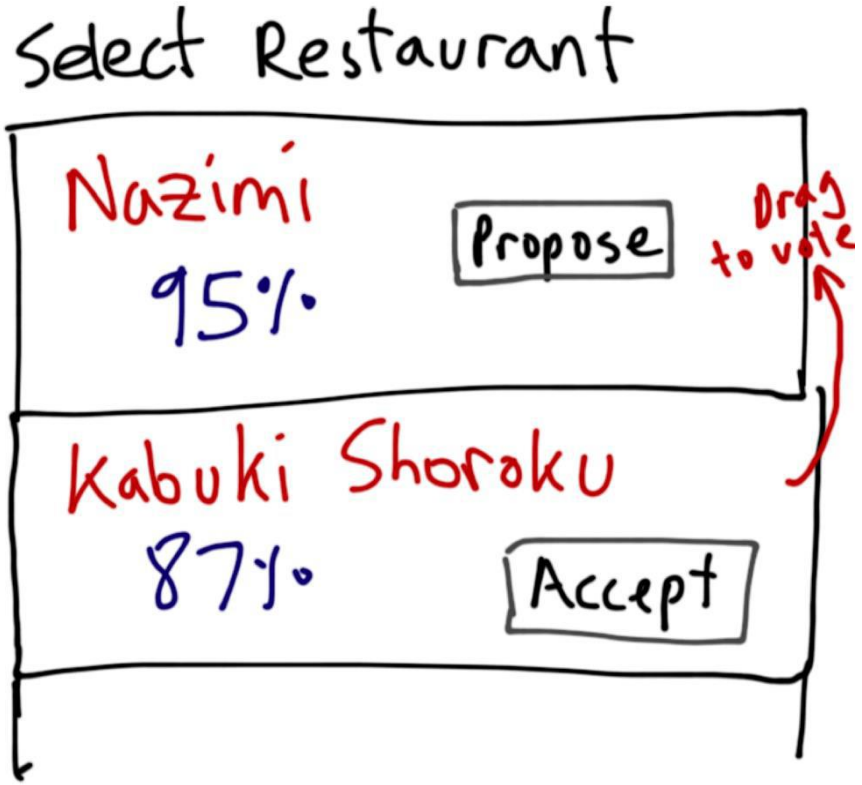
Sketching & Storyboard - Benefits

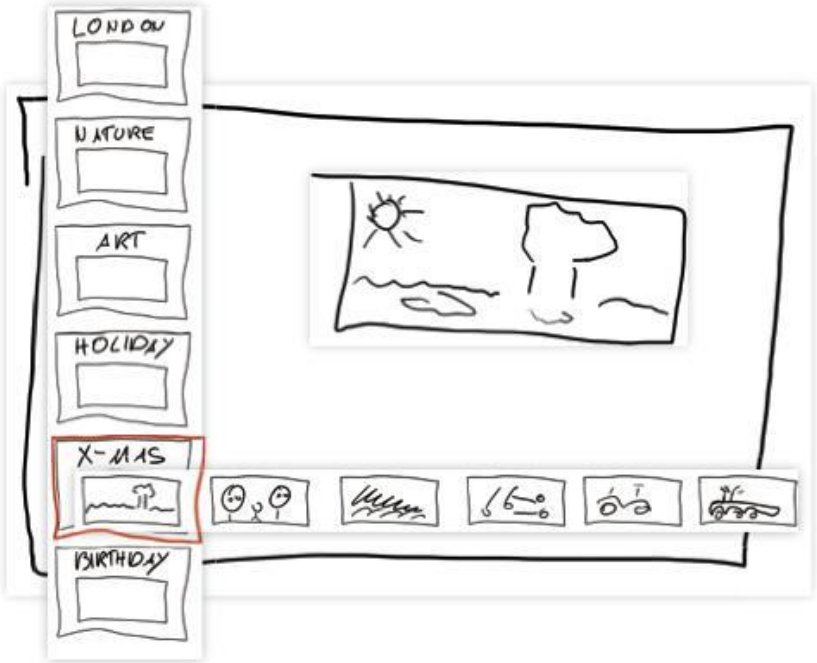
- Quick and easy to create
- Can always be drawn immediately (for a current situation)
- Cost effective: no limitation in the exploration phase
- “Disposable”: no problem drawing many sketches and discarding them again
- Clear vocabulary: a sketch implicitly conveys that it is only an idea, a first draft
- Multiple interpretation possibilities: exact meaning usually only emerges through a series of sketches/images → storyboard
- Different levels of detail (as abstract as possible to get the concept across; i.e., don’t spend time on details)

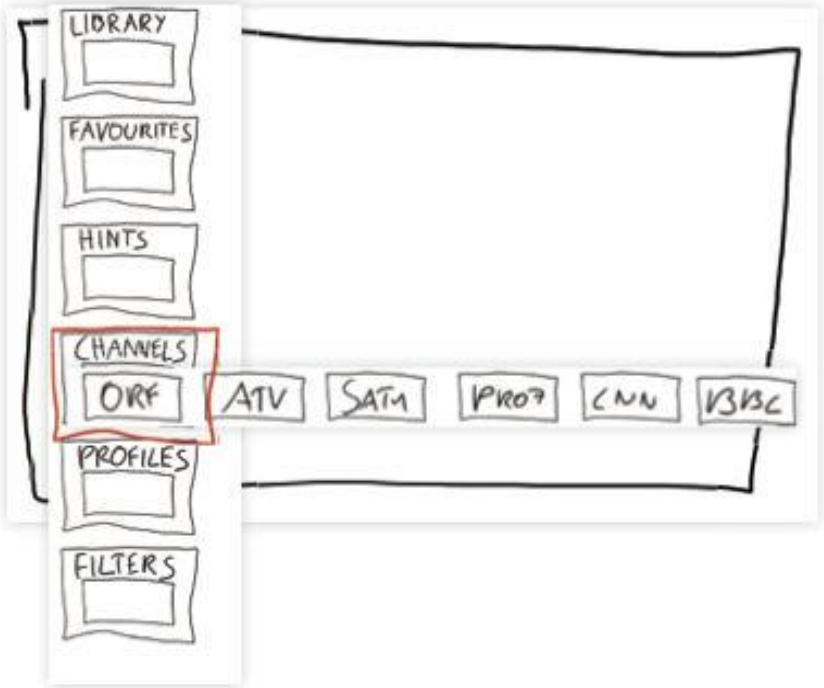
Sketching & Storyboard - Benefits

- Provide a way to develop/express/communicate ideas
- Is useful to explain (complex) content
- Think more openly and creatively
- Create abundant ideas without worrying about their quality
- Inventing concepts by quickly capturing ideas
- Capture ideas, that you come across
- Discuss, critique and share ideas with others
- Select ideas
- Archive ideas for later
- Have fun!


Choose Contact
Tracey
autocomplete
Choose Contact


Select Restaurant
Nazimi
95%
Propose
Drag to vote
Kabuki Shoroku
87%
Accept

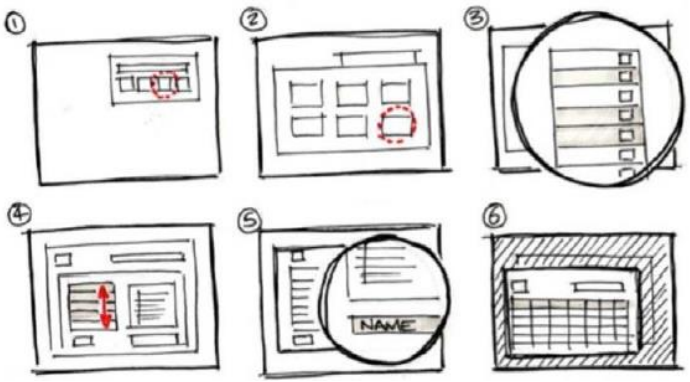

LONDON
NATURE
ART
HOLIDAY
X-MAS
BIRTHDAY

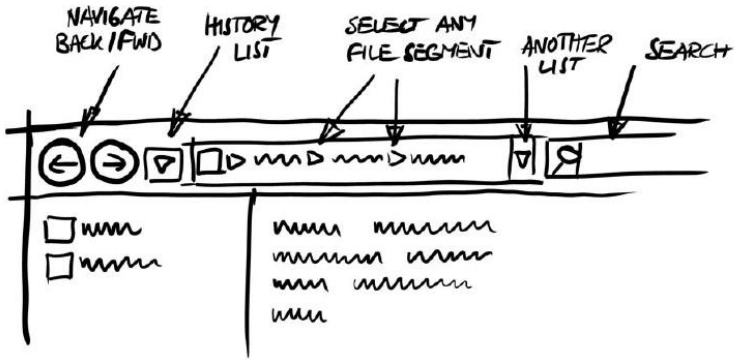

LIBRARY
FAVOURITES
HINTS
CHANNELS
ORF
ATV
SAT1
PRO7
CNN
BBC
PROFILES
FILTERS

GUI Sketching

Leave out details that are not in focus

- Focus here: navigation and search bar





Greenberg et al., 2011. Sketching User Experiences: The Workbook.

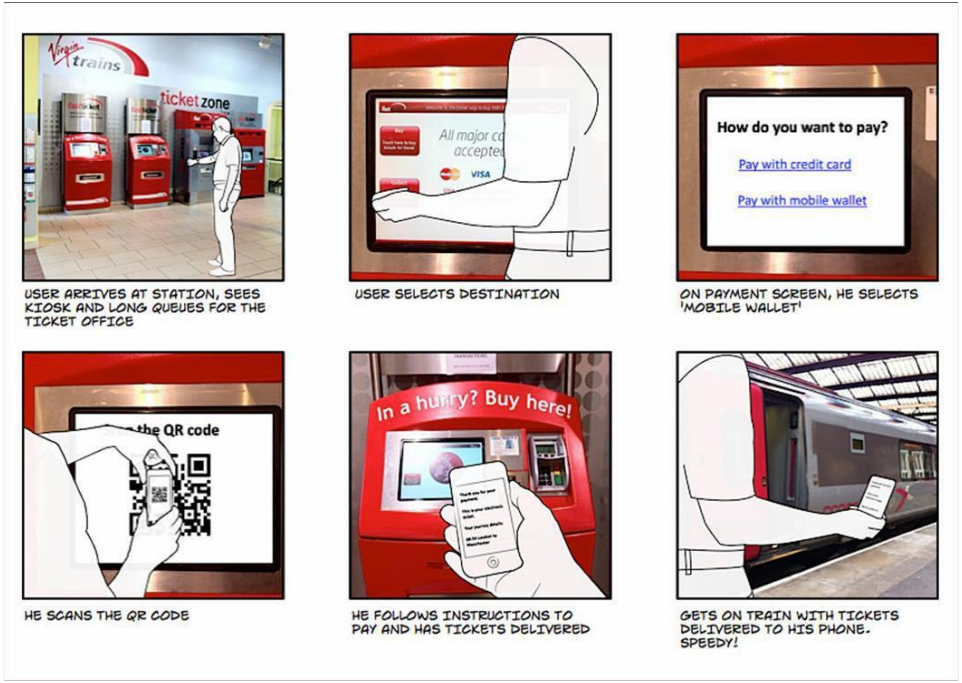
Photo Tracing

Example: Booking app for train tickets

- Combination of photos and more detailed drawings (overlay); partly created digitally
- Elaboration of details (with Adobe Photoshop, Illustrator)

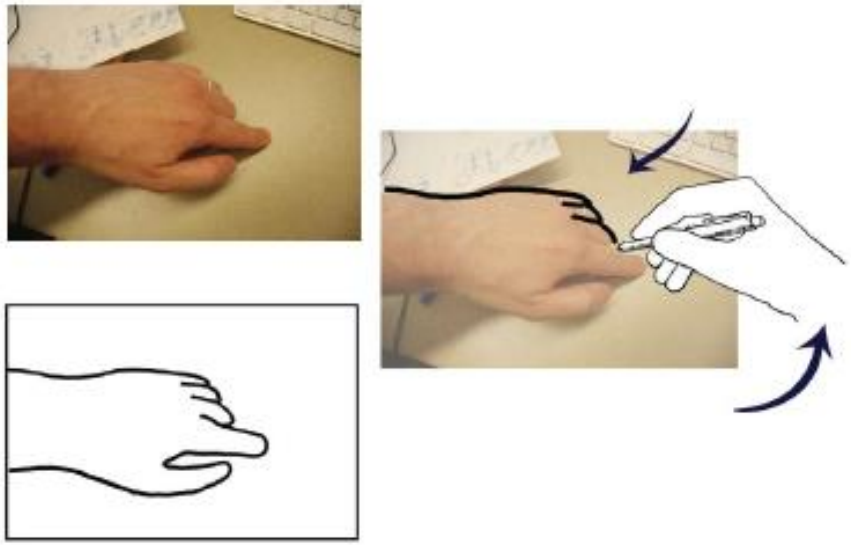
Disadvantage

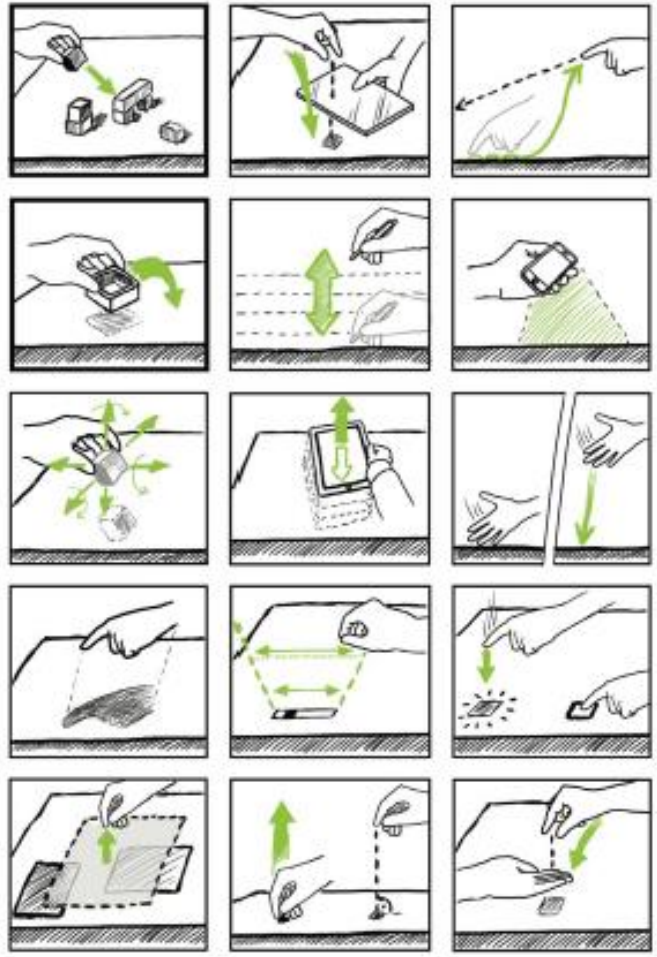
- Significantly more time-consuming (time ≈ costs)
- Only useful if presentation for third parties planned, e.g. own management, Customers, Marketing



Greenberg et al., 2011. Sketching User Experiences: The Workbook.

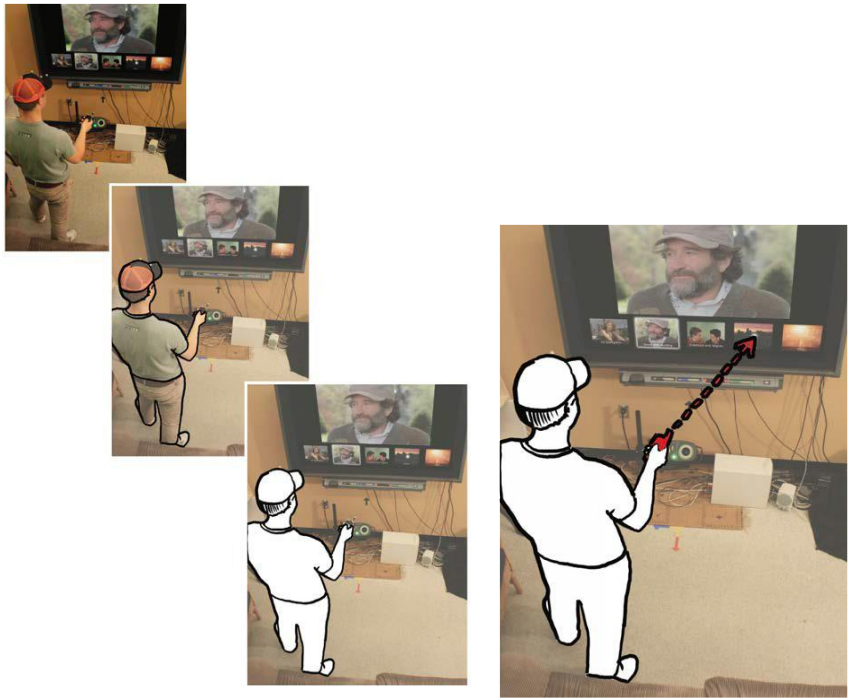
Photo Tracing



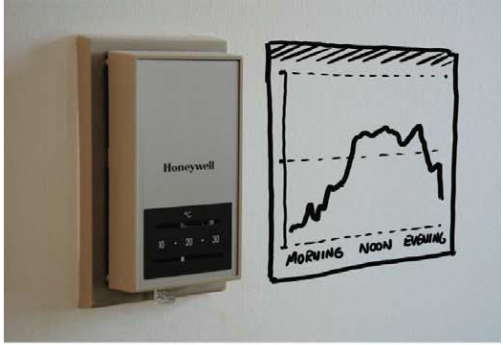


Greenberg et al., 2011. Sketching User Experiences: The Workbook.

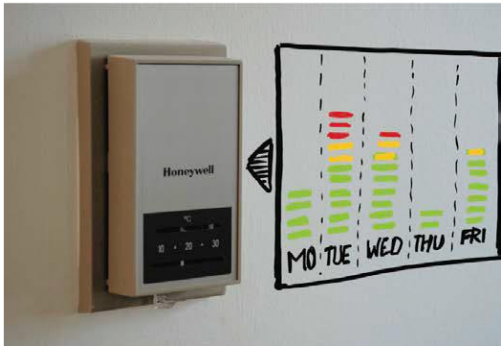
Hybrid Sketches



Augmented-Reality Thermostat Visualization:
Hybrid sketches can also help to explore different design ideas within a given context







By making the person, the phone and the annotation appear as a sketch atop the photo, the viewer can easily grasp the context of the scene, and the interaction techniques the sketch is supposed to illustrate

Greenberg et al., 2011. Sketching User Experiences: The Workbook.

Narrative Storyboard



- visual narrative: provide empathy and communicate context
- shows how persona uses the system to achieve goals
- focus on user experience, interface only part of the story
- includes context of use
- (speech) bubbles from comics can be used





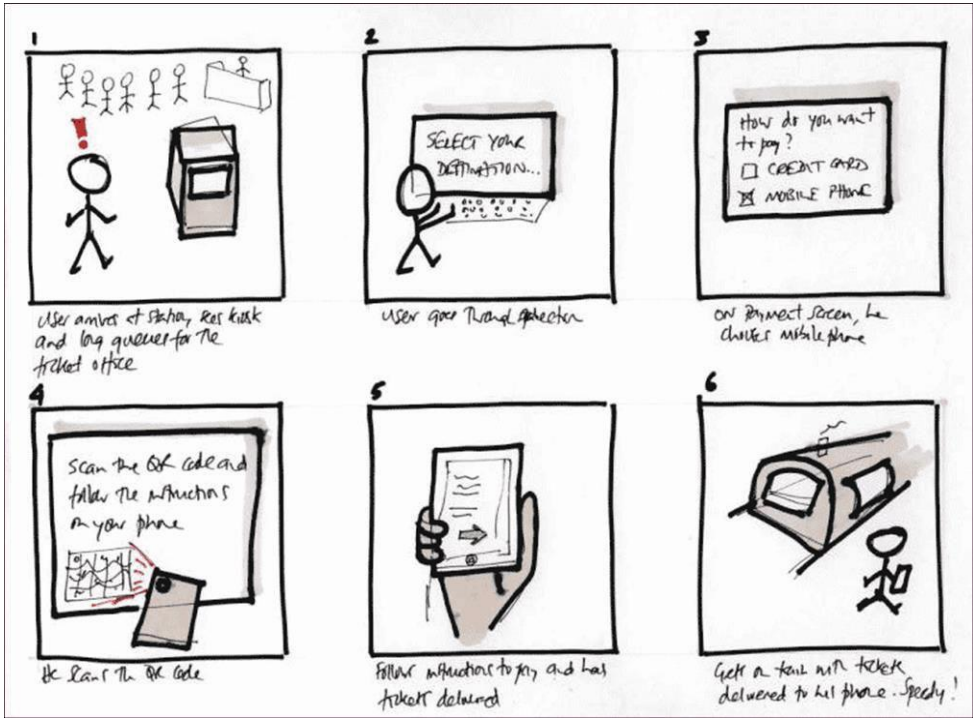




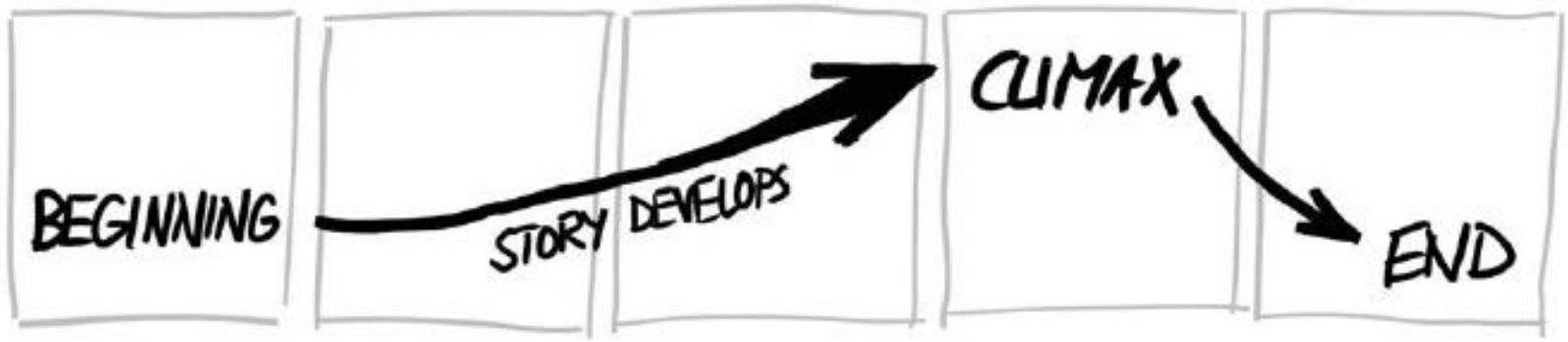
Greenberg et al., 2011. Sketching User Experiences: The Workbook.

Storyboard

- Rough sketch with annotations quickly created
- well suited to create different variants, e. g., to sketch
  - dynamic processes
  - user guidance
- decision for one version at the end...



Develop the Storyline



- Where does the interaction take place?
- What is the problem?
- What is the task a person is trying to do?
- Which people are present / What are their actions?
- What objects/devices do they use?
- What are the input/outputs?
- How do their actions solve the problem?

Greenberg et al., 2011. Sketching User Experiences: The Workbook.

Vocabulary of Camera Shots



Extreme long shot (wide shot) A view showing details of the setting, location, etc.



Long shot Showing the full height of a person



Medium shot Show's a person's head and shoulders



Over-the-shoulder shot Looking over the shoulder of a person



Point of view shot (POV) Seeing everything that a person sees themselves



Close-up such as showing details of a user interface a device the person is holding

Greenberg et al., 2011. Sketching User Experiences: The Workbook.

Devlop the Storyline



The introduction
location where interaction takes places, people involved, setting

Extreme long shot
(emphasizes context)

Over the shoulder
(emphasizes person looking at something of interest)

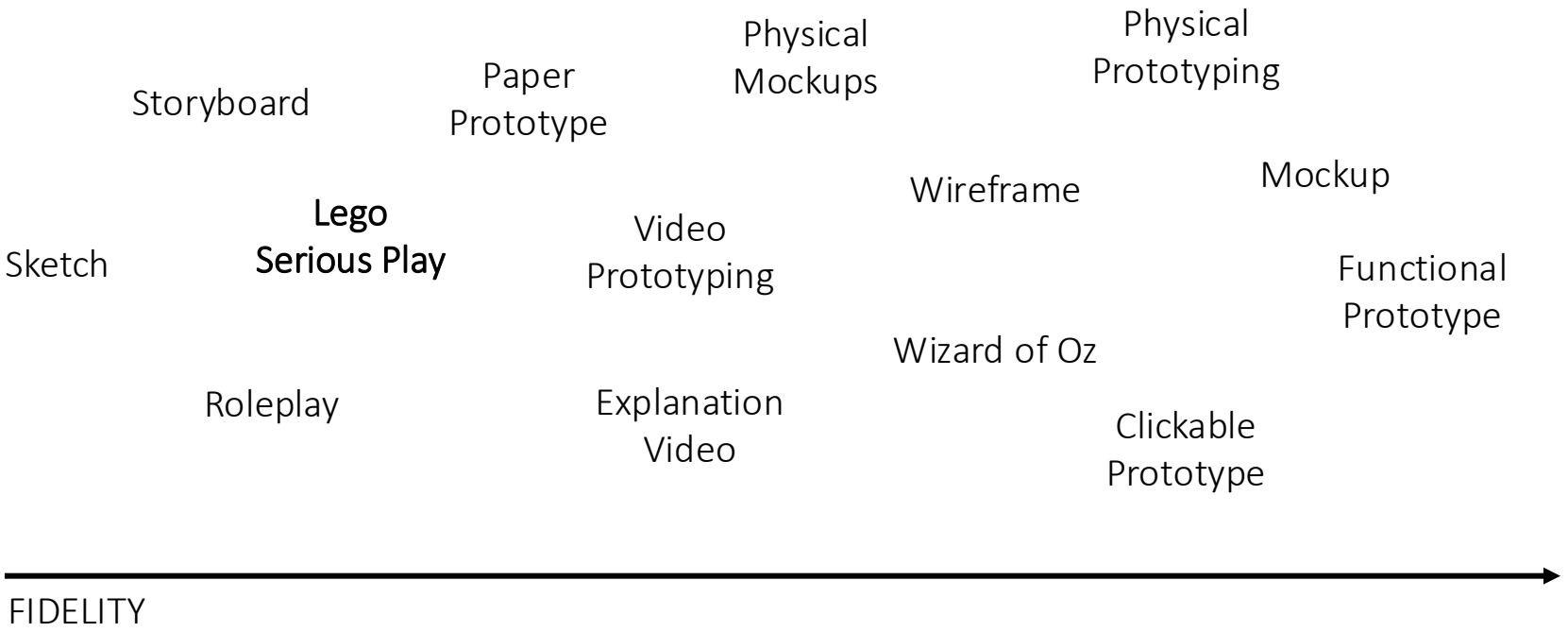
Point of view
(emphasizes person's action)

Climax
Close-up (shows details)

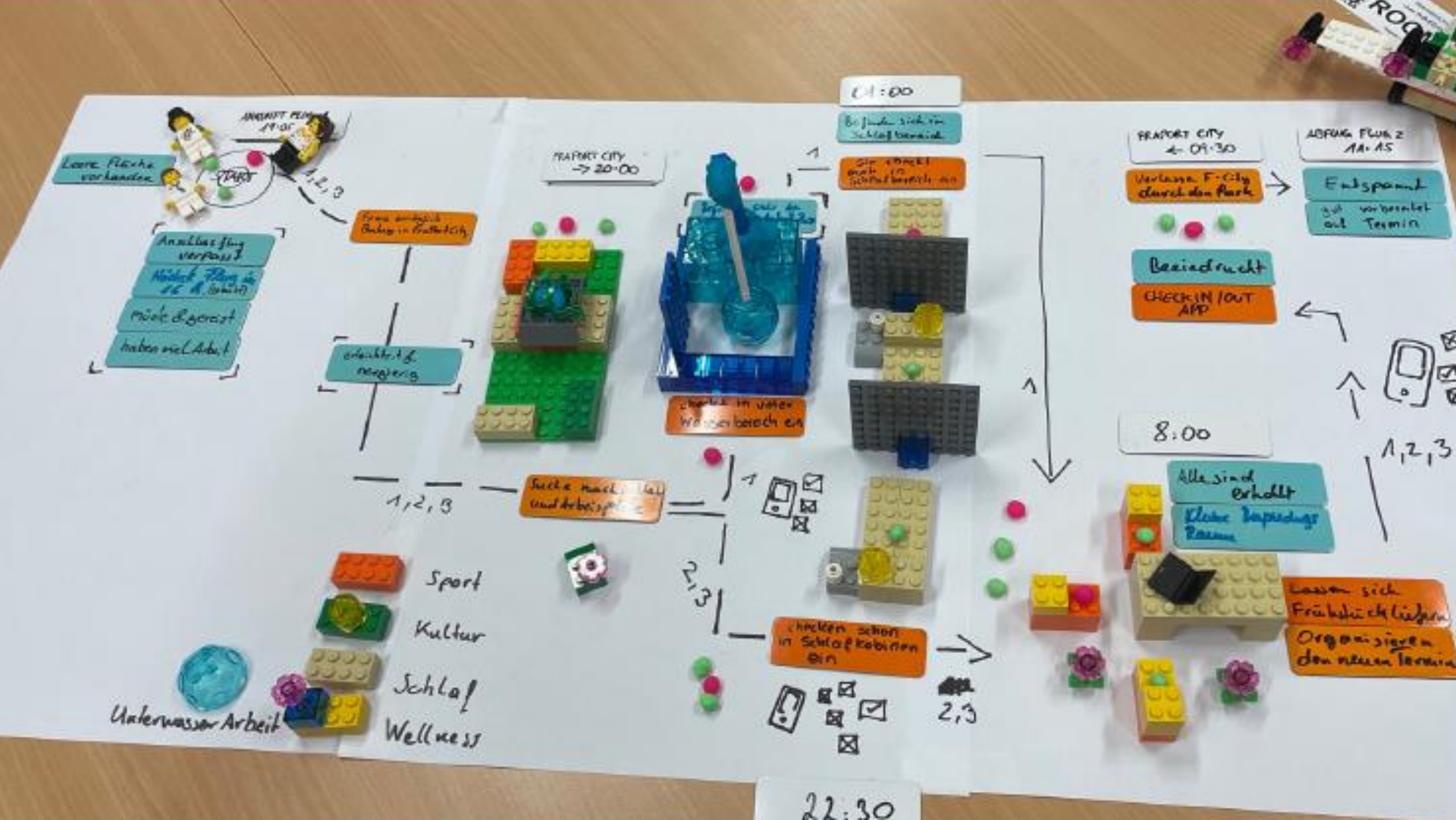
Conclusion
Extreme long shot (shows context on completion)

Greenberg et al., 2011. Sketching User Experiences: The Workbook.

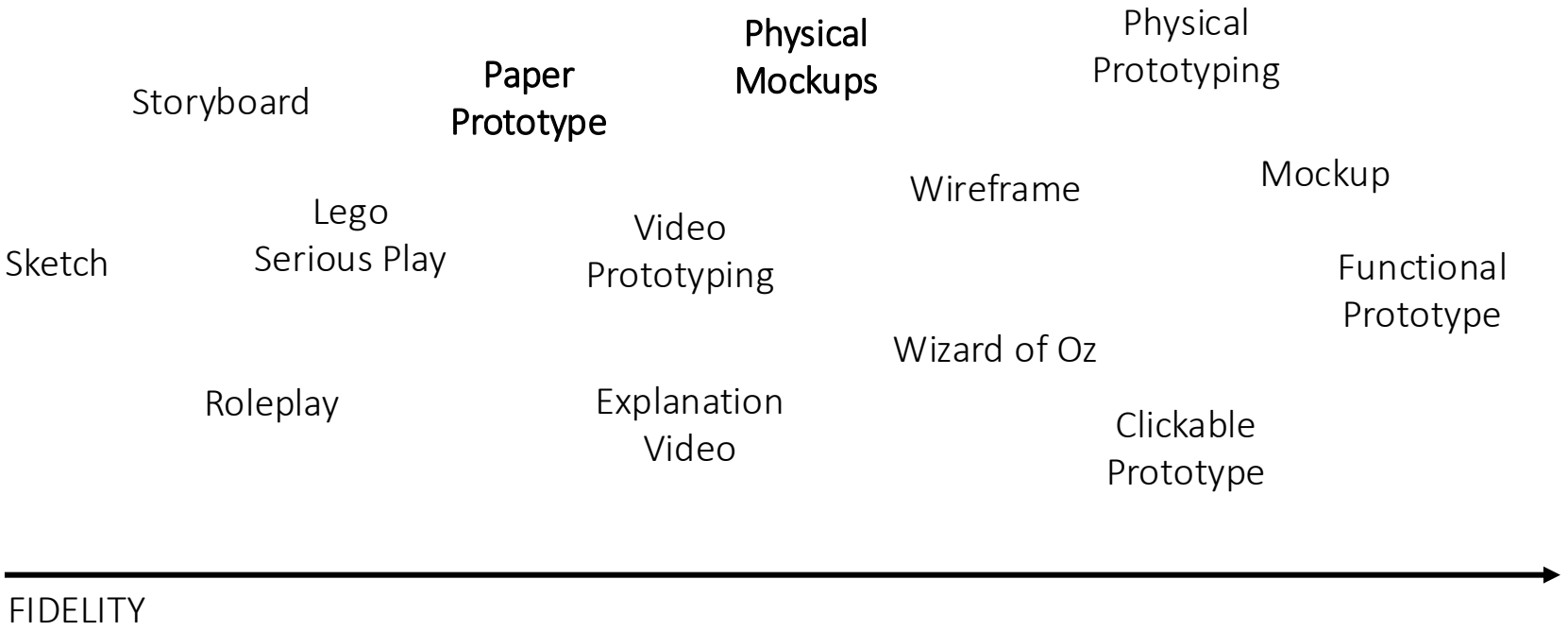
Prototyping Methods




Spediteur
Lubo
4B
Luboš scheitert bei der Anmeldung
Luboš freut sich über die In-time übersetzung durch das device
38
Luboš ist erstaunt über die tolle Selbstanmeldu


Leere Fläche vorhanden
START
1,2,3
Anreise/Flug verpasst
Hotel/Flug ab 16 h (Gebiet)
Müde & gereizt
haben viel Arbeit
Frau ... Ankunft in Fraport City
erschöpft & mürrisch
1,2,3
Suche nach ... und Arbeitsplatz
Fraport City -> 20:00
checken in ... Wasserbereich ein
11:00
Sie checkt auch im ... Bereich ein
1
1
1
2,3
checken schon in Schlafkabinen ein
2,3
Sport
Kultur
Schlaf
Wellness
Unterwasser Arbeit
Fraport City ← 09:30
Verlasse F-City durch den Park
Abflug Flug 2 11:15
Entspannt
... vorbereitet auf Termin
Bezahlt
CHECK IN/OUT APP
8:00
Alle sind erhalten
Kleine Impfung Pause
1,2,3
Lassen sich Frühstück liefern
Organisieren den neuen Termin
22:30

Prototyping Methods





Paper Prototyp

- the structure of paper prototypes is more defined than that of set of sketches
- typically involve using stencils and cardboard to create more substantial, detailed mockups of various Web pages or application screens for use during usability testing
- similar to sketching, paper prototyping is often collaborative and is a good first step before moving on to an interactive prototype

Christian Moser: User Experience Design: Mit erlebniszentrierter Softwareentwicklung zu Produkten, die begeistern, Springer Verlag, 2012

Paper Prototyp

Advantages

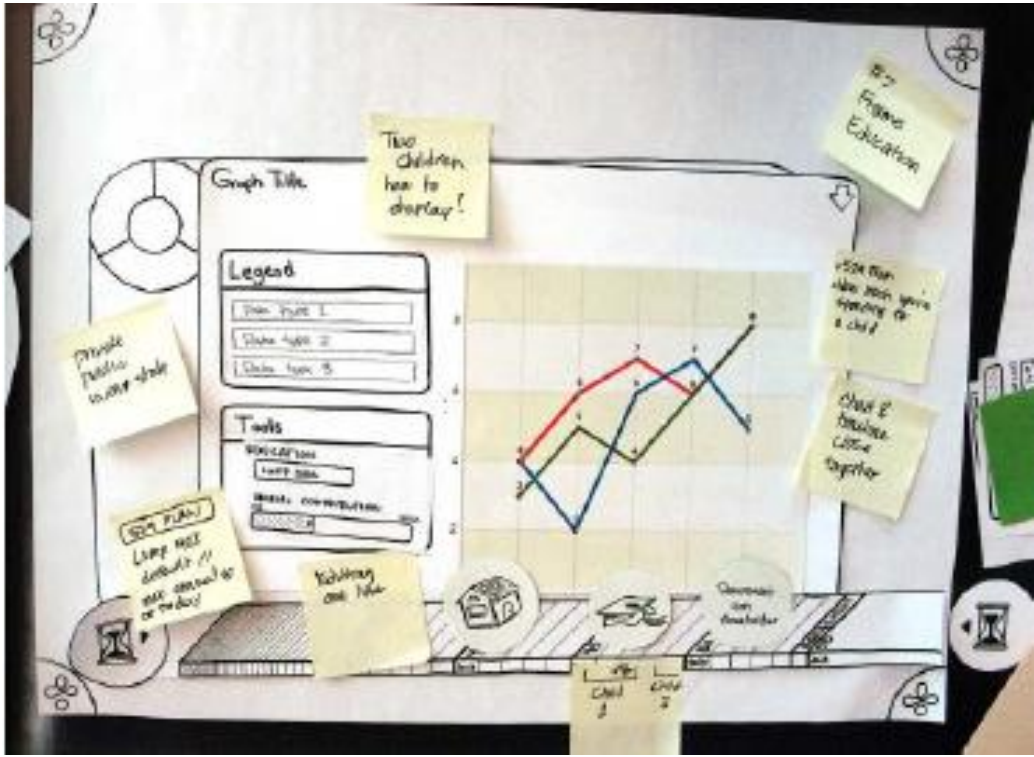
- is fast, easy, inexpensive, and requires no technical knowledge
- enables early user testing
- quick creation of different variations
- use of realistic dimensions (e.g. tablet & smartphone size)
- critical design flaws can be detected and corrected at an early stage



Paper Prototyp

Disadvantages

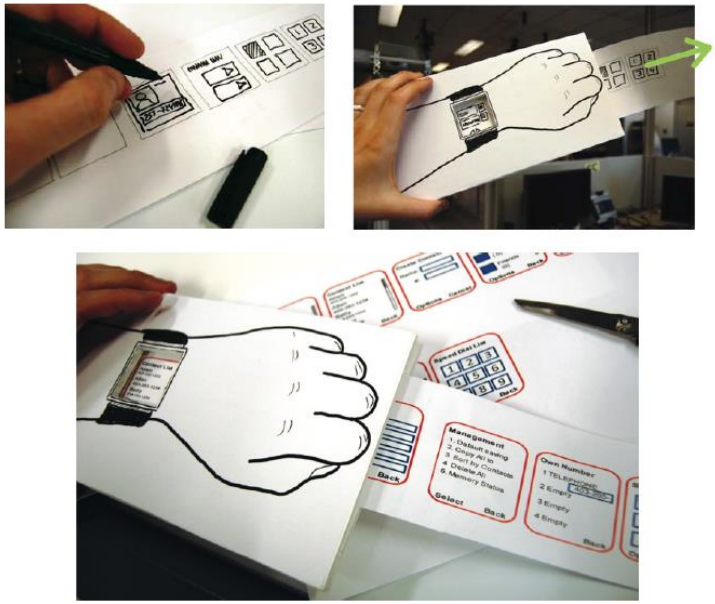
- Less realistic and cannot find all usability issues
- Cannot evaluate performance (input, etc.)
- No remote testing

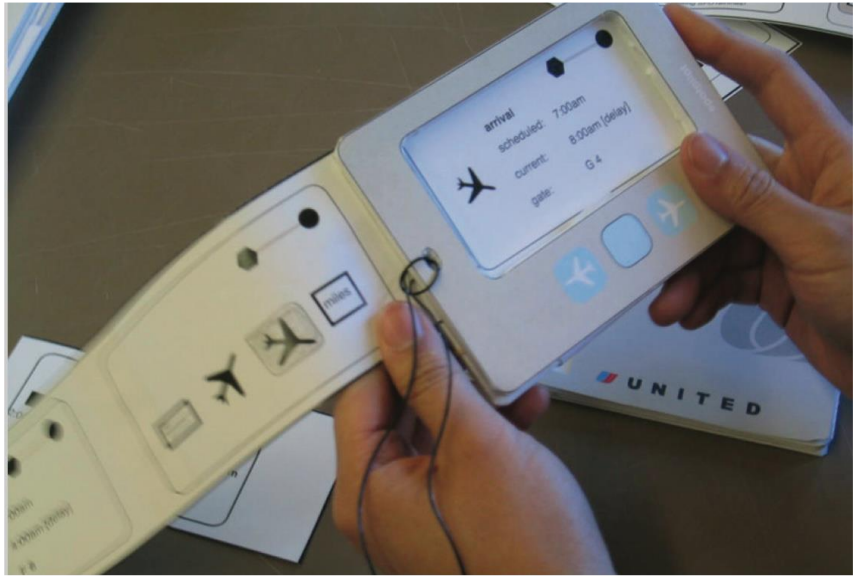



ABOUT
SKILLS
SKILLS
BOOK SESSION
DATE
TIME
SESSION TYPE
NOTES:
THANKS

Foam core prototypes

Sketching a novel interface for a digital watch



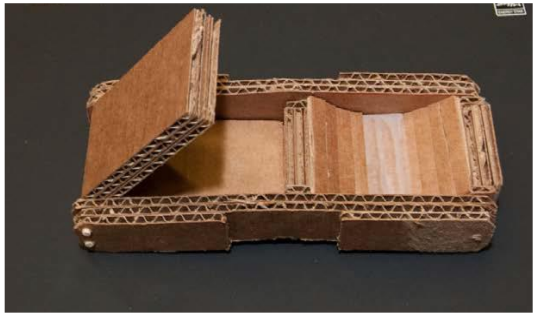


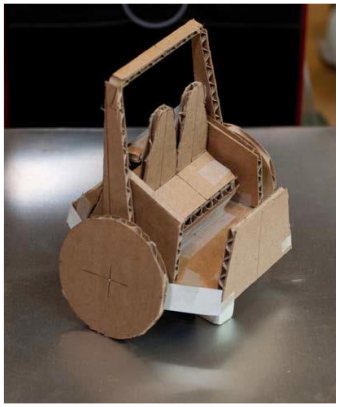
Foam core prototype of a hand-held device

S. Greenberg, S. Carpendale, N. Marquardt, B. Buxton: Sketching User Experiences - The Workbook, Morgan Kaufmann, 2011

Physical Mockups

a physical mockup is an embodied sketch because it is even more of a physical manifestation of a design idea and it is a tangible artifact for touching, holding, and acting out usage







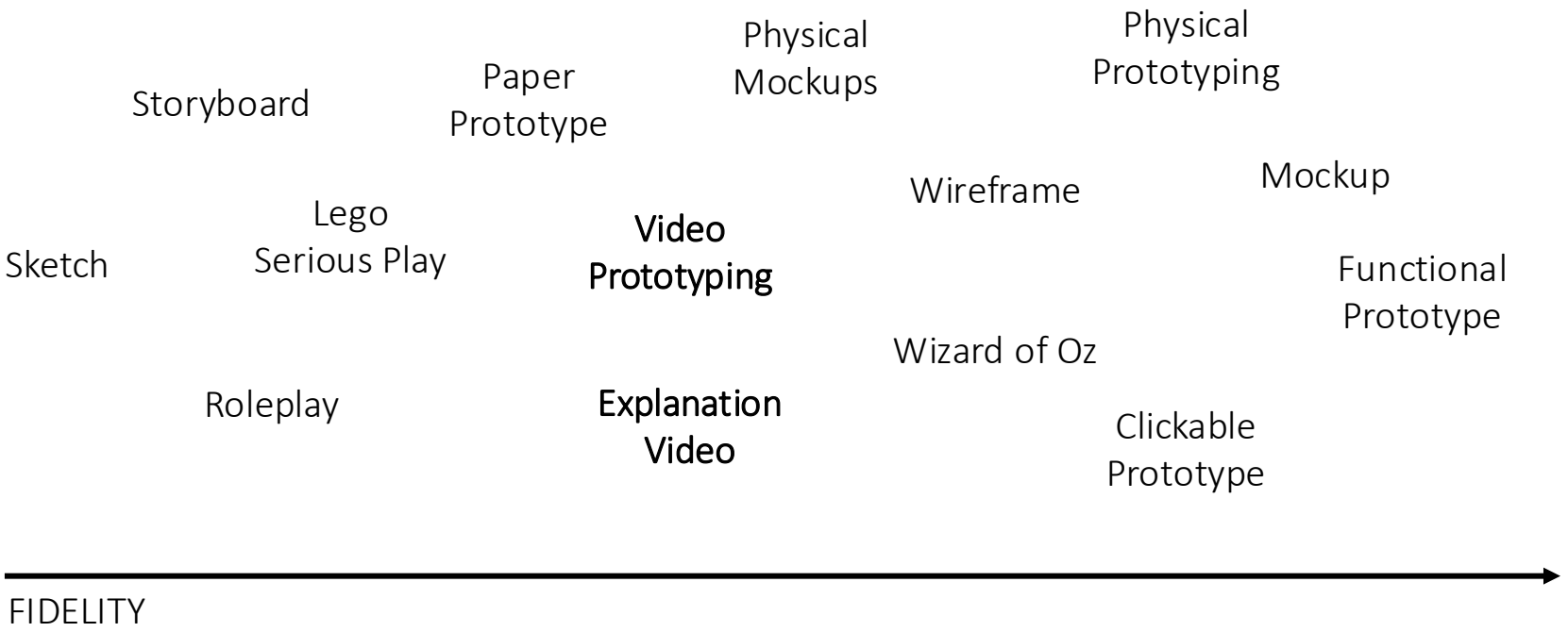
Example of a rough physical mockup
Akshay Sharma, of the Virginia Tech Department of Industrial Design

S. Greenberg, S. Carpendale, N. Marquardt, B. Buxton: Sketching User Experiences - The Workbook, Morgan Kaufmann, 2011




Latte Macchiato
on/off
milk
ml
EQ.9
s 700
SIEMENS

Prototyping Methods



Video Prototyp

Ideas and concepts may also be visualized interactively, in terms of videos or animations:

- Explanation Video
  demonstrates the overall flow, similar to storyboards

- Video Prototype
  shows the interaction in a real usage scenario
  - Animated paper prototype
  - Screen capture/screencast
  - Computer animations
  - Doll-house style with actors / real actors

Stop Motion

- illustrates how users can interact with the system
- gives an impression of which gestures are suitable for different devices
- a variety of low-cost special effects are possible





https://www.youtube.com/watch?v=ep-kLD59uVY

Video Prototypes

Advantages

- Can be used in early development phase
- Allows to easily demonstrate dynamic interactions
- No experts (developers, etc.) needed

Disadvantages

- Some knowledge about video recording/cutting is needed
- No “real” interaction possible (just watching the video)
- Does not allow a “real” evaluation (A/B testing with statistics)

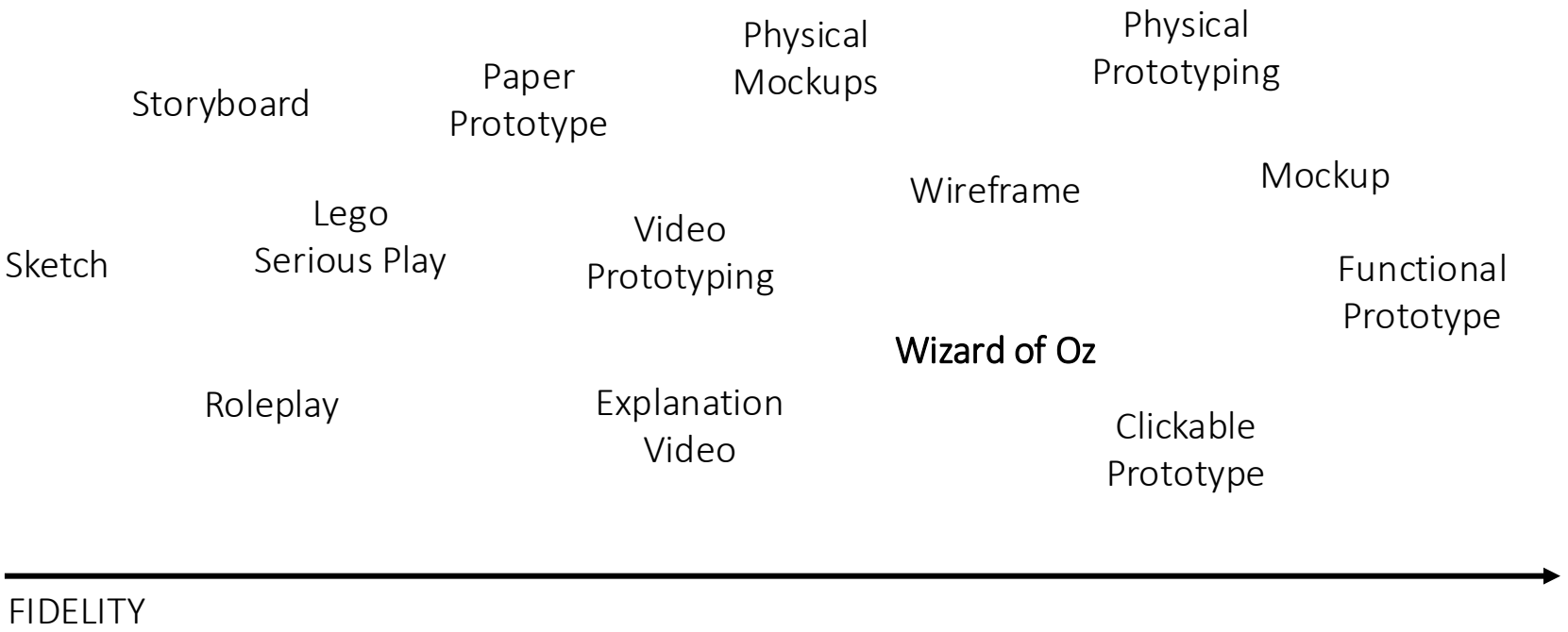
Proof of Concept Video Technique

Fictional video that shows a „working“ system in a variety of scenarios



https://www.youtube.com/watch?v=6TpsWOsL96U

Prototyping Methods

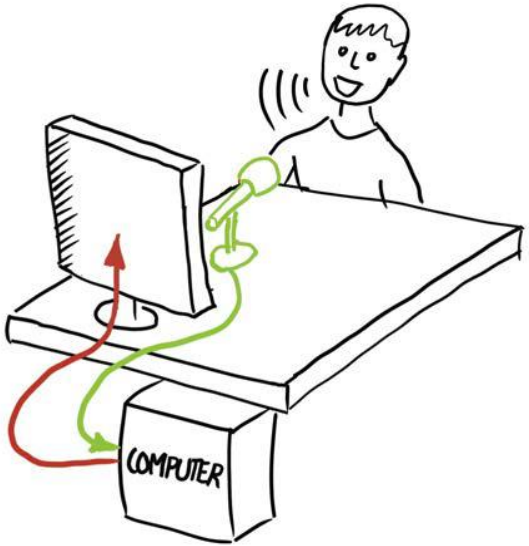


Wizard of Oz

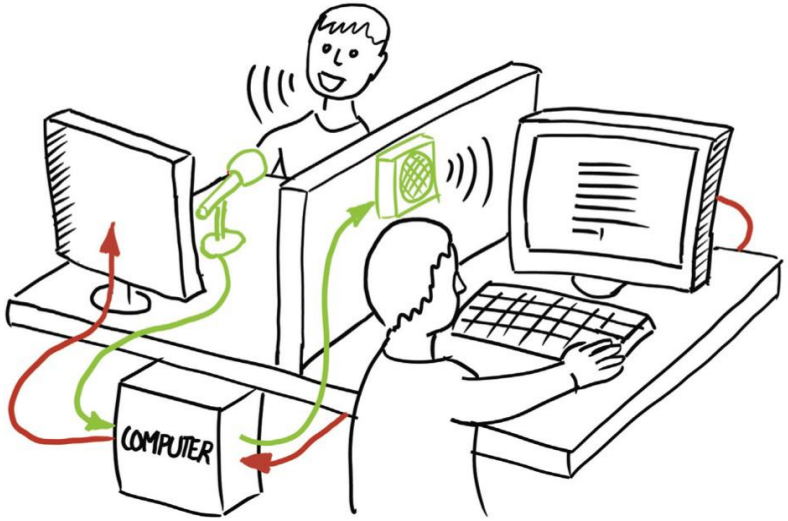
- used to model functionality of a product by having a human operator (wizard) perform the task usually performed by the compute
- The user has the impression that the system is working
- Based on the novel “The wonderful wizard of Oz” (Lyman Frank Baum, 1900)
- In contrast to paper prototypes, the wizard is hidden (may require fast reaction)
  - i.e., simulating an AI chat dialog partner (wizard has to type fast)
  - Wizard can simulate errors on purpose

S. Greenberg, S. Carpendale, N. Marquardt, B. Buxton: Sketching User Experiences - The Workbook, Morgan Kaufmann, 2011

Wizard of Oz



What the user saw



What was actually happening

S. Greenberg, S. Carpendale, N. Marquardt, B. Buxton: Sketching User Experiences - The Workbook, Morgan Kaufmann, 2011

Wizard of Oz

- when design concepts involve intelligent or adaptive responses that cannot be effectively conveyed in mockups
- when design concepts include sophisticated non-screenbased interactions
- when designs being tested are beyond what current technology can do
- when design ideas are still wide open and you want to see how users behave naturally in the course of simulated interaction



Testing an Alexa-Skill



Wizard of Oz

Advantages

- save time and money when compared to building functional systems
- creating multiple variations is easy
- places the user at the center of development
- can envision challenging-to-build applications

Disadvantages

- wizards require training and can be inconsistent
- user may have unrealistic expectations for the system

http://crowd.com.au/test/

Examples

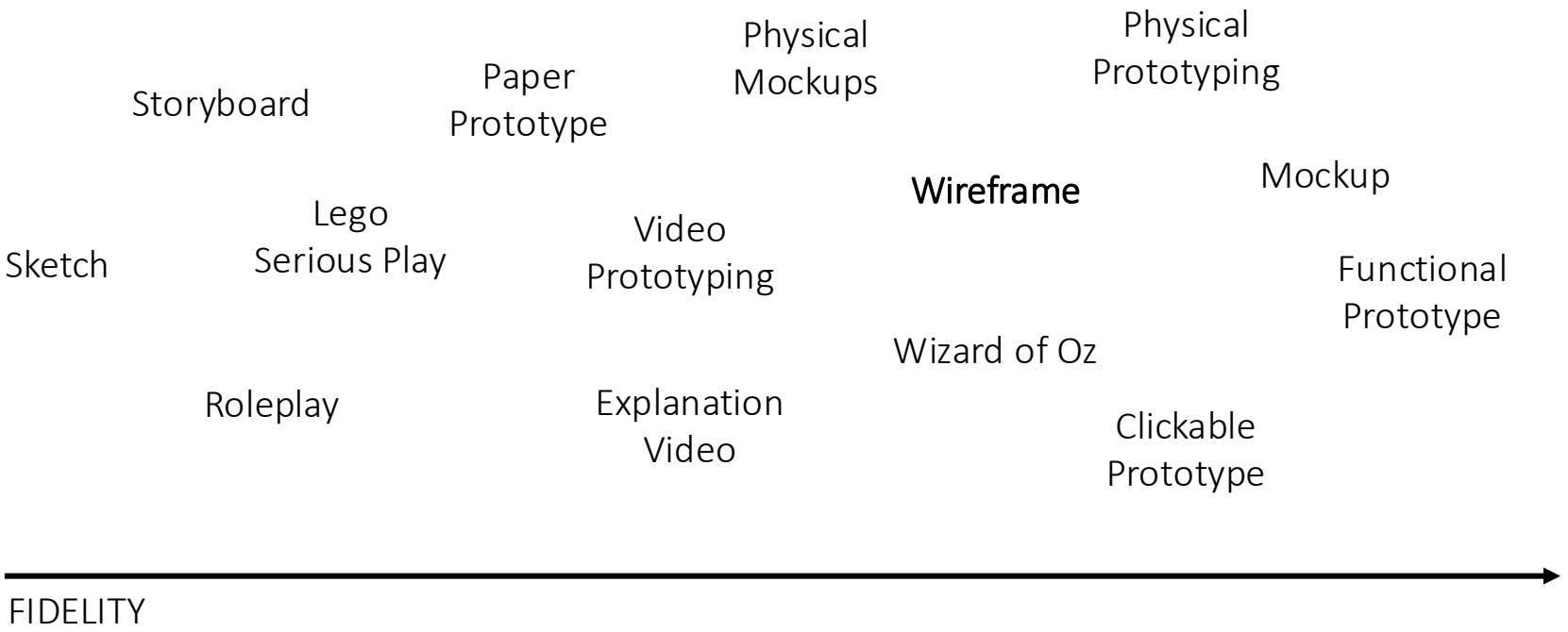
“Ghost driver”:
https://www.youtube.com/watch?v=X7Fkmf0XHiY

“CardBot”:
https://www.youtube.com/watch?v=Z3fq0UJL7Kw

“Interactive Wall”:
https://www.youtube.com/watch?v=vetw8sE8I_s



Prototyping Methods

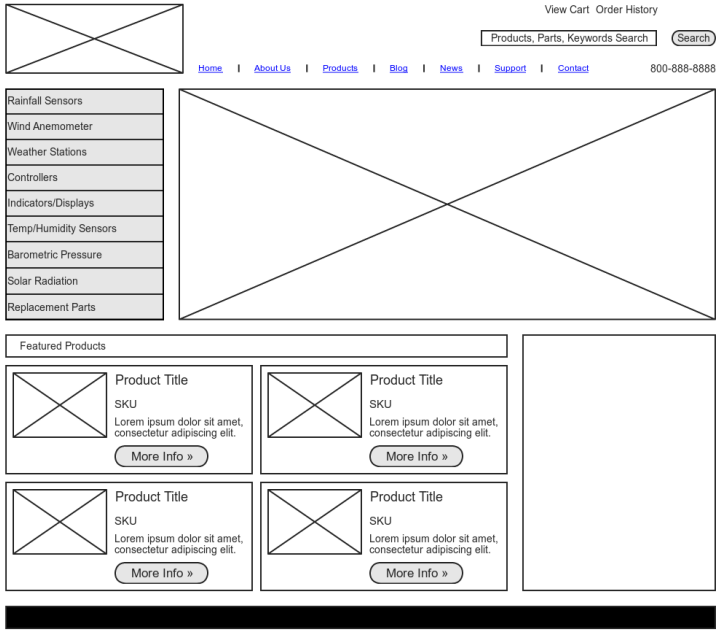


Wireframe

- a visual representation screen of an application that shows basic layout and navigation elements as wireframe
- focus on structure of user interface, no details like color and images

Wireframes are basic building blocks of UX prototypes

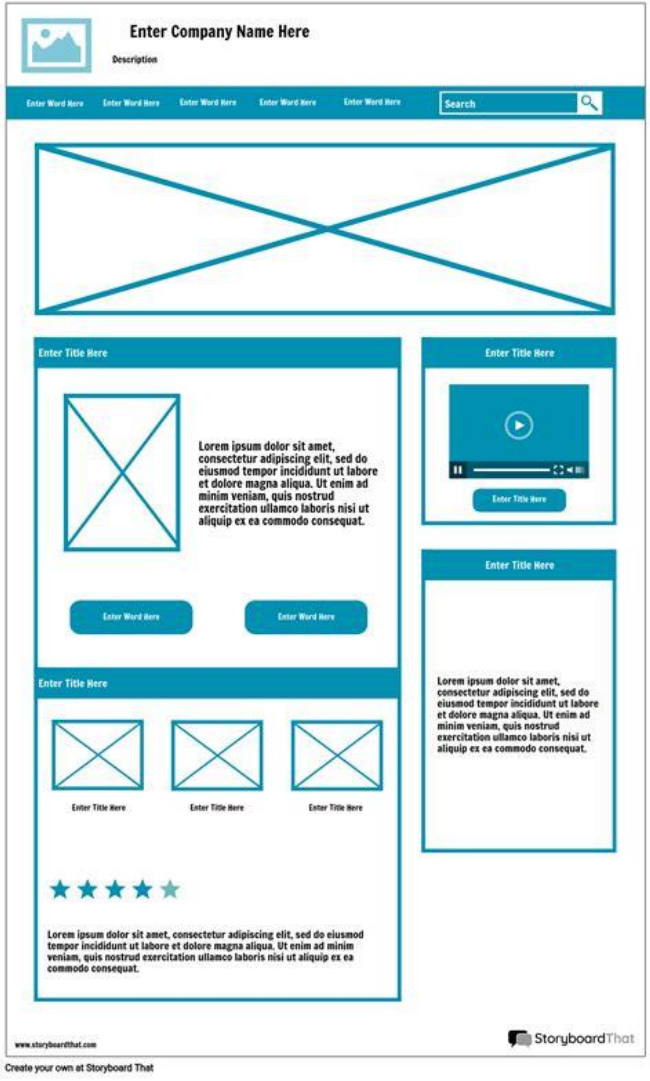
- they can test early concepts of functionality, presentation of interactive components, and layout
- they support gathering of feedback before full system has been thought through
- they are fast to create and cheap to discard

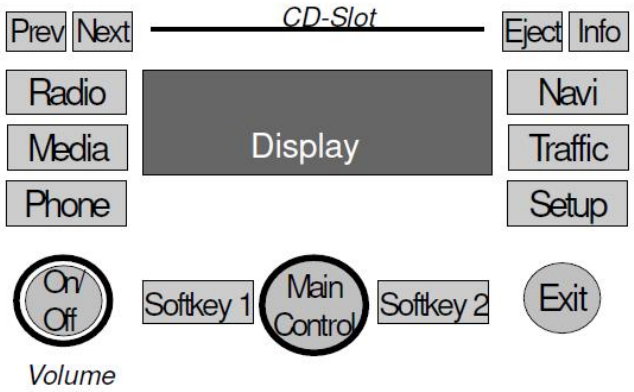


Christian Moser: User Experience Design: Mit erlebniszentrierter Softwareentwicklung zu Produkten, die begeistern, Springer Verlag, 2012

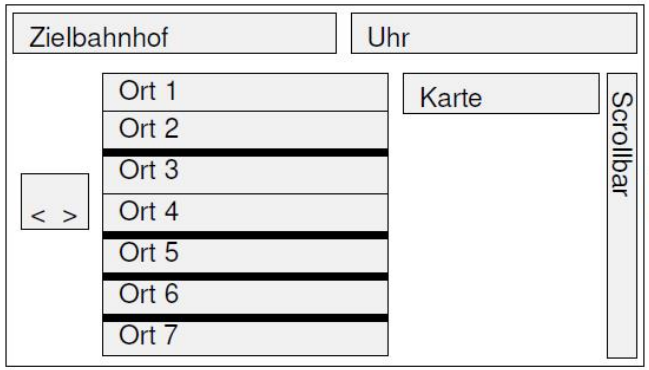
Wireframe

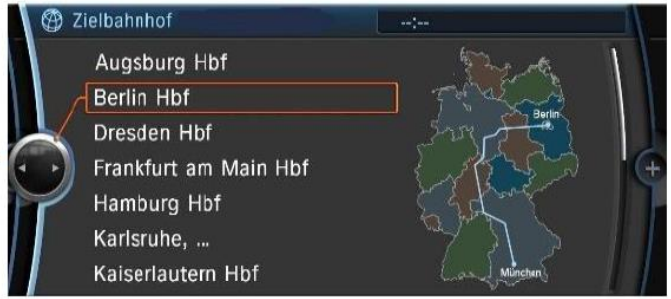
- allows to demonstrate a “skeleton” of the user interface
- size, position, relations, and (partly) also design represented as planned
- content doesn't have to be final, but it is recommended to use real texts instead of dummy texts to get a feeling for needed space




Prev
Next
CD-Slot
Eject
Info
Radio
Media
Phone
Display
Navi
Traffic
Setup
On/ Off
Softkey 1
Main Control
Softkey 2
Exit
Volume

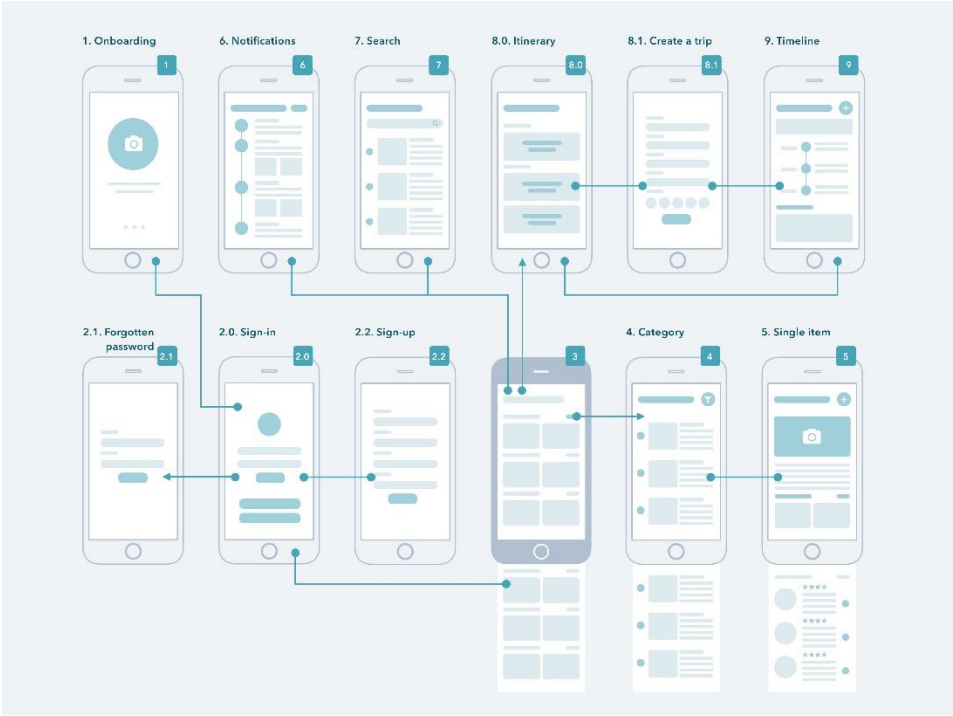

AMUNDSEN
MP3
RADIO
MEDIA
PHONE
NAV
TRAFFIC
SETUP
19:02
STARA BOLESLAV
50m
610
462km
E65
23:25


Zielbahnhof
Uhr
Ort 1
Ort 2
Ort 3
Ort 4
Ort 5
Ort 6
Ort 7
Karte
Scrollbar
< >


Zielbahnhof
Augsburg Hbf
Berlin Hbf
Dresden Hbf
Frankfurt am Main Hbf
Hamburg Hbf
Karlsruhe, ...
Kaiserlautern Hbf
Berlin
München

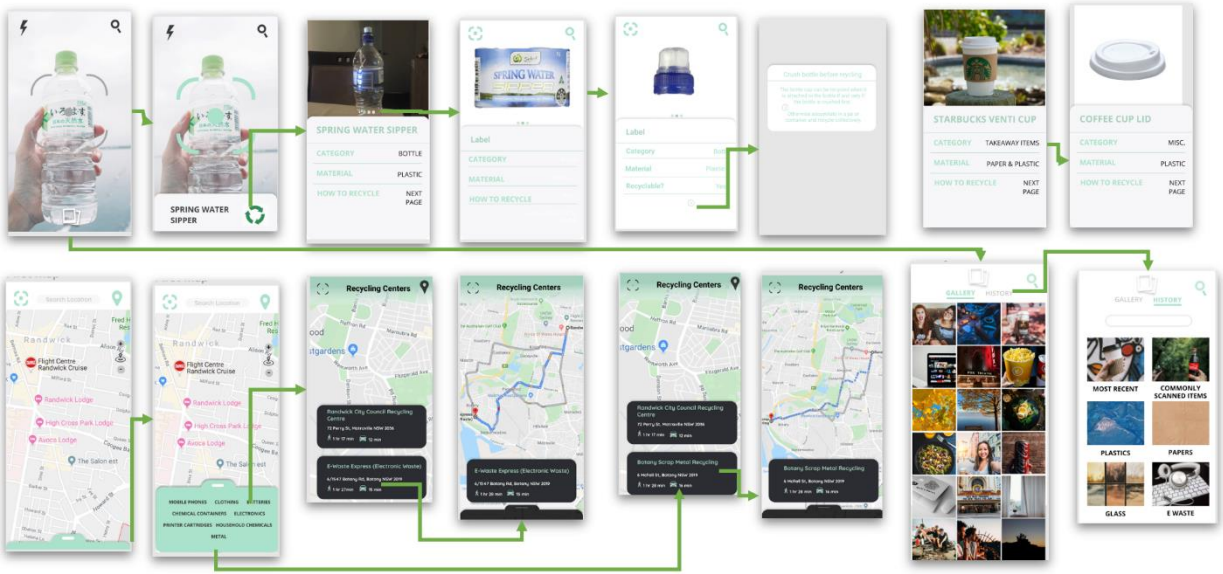
Wireflow

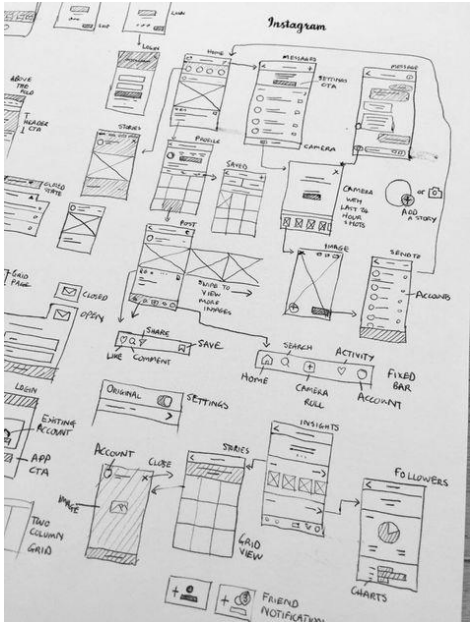
- even though in industry practice, the plural form of the term wireframe is used to denote flows and sequences of individual **wireframes**, the more accurate term is a **wireflow**
- illustrates navigational flow within an interaction design
- nodes are wireframes
- arcs are arrows representing navigation flow among the wireframe



Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019

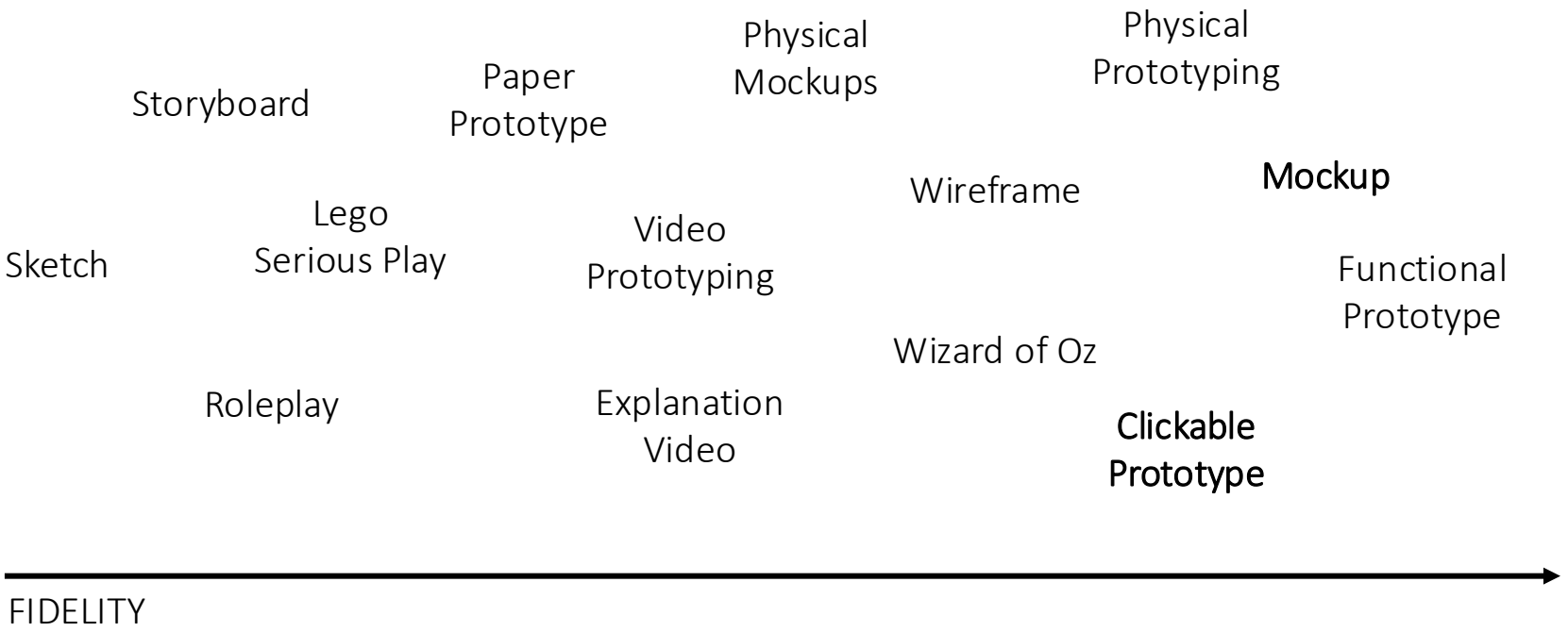
Wireflow





Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019

Prototyping Methods



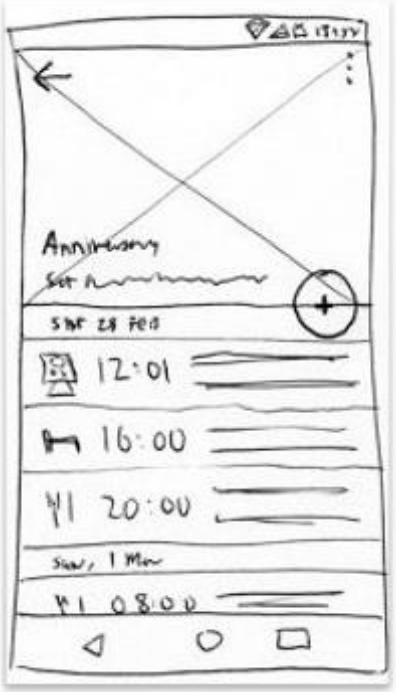
Wireframe vs. Mockup

Wireframe (low-fidelity)

- basic structure and functionality
- breakdown into individual screens

Mockup (high-fidelity)

- visual look-and-feel
- show actual content
- use of colors, typograph, images, icons, etc.
- how should the interface look like?





https://mentormate.com/blog/low-fidelity-wireframes-vs-high-fidelity-wireframes

Course - 1

Courses

Our most popular categories

Coding

Cooking

Browse our newest courses

Mathematics for Machine Learning

Peter Müller

Introduction to C#

Lisa Meier

Our recommendations for you

Course - 2

Mathematics for Machine Learning

Coding | Machine Learning | Software

What you will learn

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin pellentesque, leo vel varius vulputate, lacus elit viverra ante, vel malesuada libero dolor eget purus. Cras molestie, nisl vitae blandit malesuada, sem est facilisis leo, sollicitudin commodo felis ex eu nibh. Sed fringilla rutrum est ac luctus. Mauris non nisi ultricies, consectetur nisi at, mattis tortor. Integer ornare, diam et tincidunt sodales, odio mauris blandit ex, non mattis est sem posuere tellus. Nam rutrum suscipit nulla at pretium. Aliquam non neque non sapien varius feugiat.

Who this course is for

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin pellentesque, leo vel varius vulputate, lacus elit viverra ante, vel malesuada libero dolor eget purus. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Available dates

20.02.2020 08:00-17:00 Uhr >

27.02.2020 08:00-17:00 Uhr >

04.03.2020 08:00-17:00 Uhr >

11.03.2020 08:00-17:00 Uhr >

Course - 3

Mathematics for Machine Learning

Selected Date & Location

Monday, 20.02.2020 08:00-17:00 Uhr

Vienna, Lasallestraße 1a



Price

€ 120

(inkl. 20% USt.)

I ACCEPT THE TERMS AND CONDITIONS.

ENROL NOW

Clickable Prototype

Paper Prototype:

- physical
- quick and easy to create
- no tool-specific knowledge required

Clickable Prototype

- digital
- easy to share and distribute
- quick change possible if you are familiar with the tool



https://www.uxpin.com/studio/blog/engaging-prototype-gif/

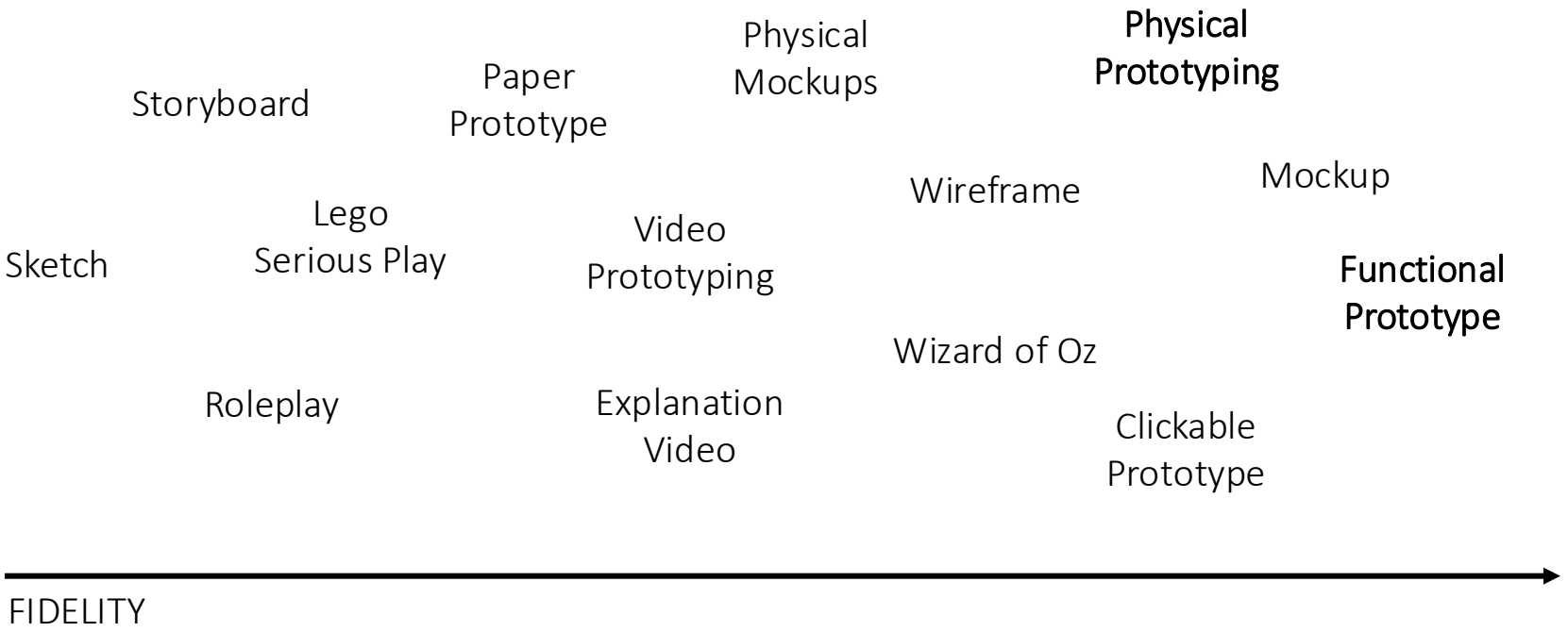
Digital Prototyping Tools

- Axure RP
- Balsamiq
- InVision
- Adobe XD
- Sketch
- Figma
- Principle
- ProtoPie
- …

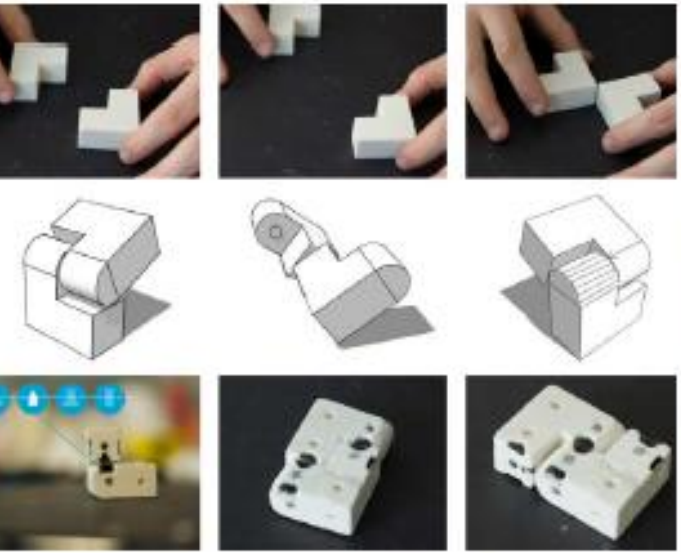


https://www.uxpin.com/studio/blog/engaging-prototype-gif/

Prototyping Methods



Tangible Prototype



Virtual and real prototypes



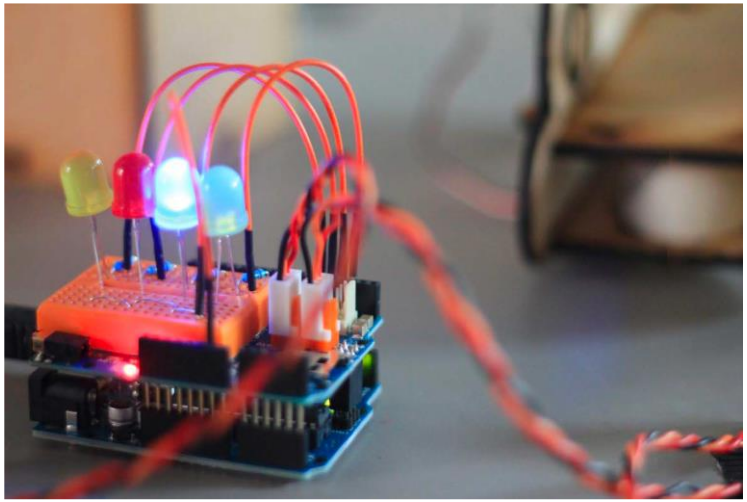
3D printer



Arduino

Coded Prototypes

physical computing kits and software development kits facilitate the transition from design to construction





Low vs. High-Fidelity Prototypes

Low-Fidelity Prototype

Advantages
- quick revision possible
- more time can be spent on improving the design before starting development
- evaluates multiple design concepts
- useful communication device

Disadvantages
- limited error checking
- poor detailed specification for development
- limited usefulness for usability tests
- navigational and flow limitations

High-Fidelity Prototype

Advantages
- (almost) complete functionality
- fully interactive
- look and feel of the intended product

Disadvantages
- more resource-intensive to develop
- time-consuming to modify
- inefficient for proof-of-concept designs

H. Sharp., J. Preece, Y. Rogers.: Interaction Design - beyond Human-Computer Interaction, Wiley; 5th Edition, 2019